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Latent Regime Discovery in Black Hole Accretion Disks via Structural Invariance

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

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June 2026.

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Declaration

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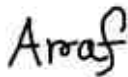
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Abstract

Modern astronomical observatories produce high-cadence, multi-wavelength data capturing the variability of black hole accretion systems, which transition between distinct physical states differing in spectral emission, accretion geometry, and jet activity. Observational datasets are, however, unlabelled: the accretion state responsible for each observation is not directly observable. Existing causal discovery frameworks require predefined environment labels, surrogate domain indices, or prior knowledge of the latent state structure. Methods that do discover regimes jointly with causal structure, notably CASTOR, rely on expectation-maximisation over the full series and provide no explicit statistical test for whether a proposed boundary reflects a genuine mechanism change rather than finite-sample variation. Furthermore, the temporal dependence present in high-cadence monitoring data violates the i.i.d. assumptions underlying many causal discovery algorithms, inflating test statistics and producing spurious boundaries.

We propose **LIRA** (Latent Invariant Regime Analysis), a causal artificial intelligence framework for discovering hidden causal regimes and regime-specific causal structures directly from unlabelled time-series observations. The method is motivated by the power-law physics of the Fundamental Plane of Black Hole Activity, which justifies modelling observational variables within a log-linear structural causal framework. Regime discovery proceeds by recursive binary partitioning guided by a Chow-test invariance criterion, with temporal autocorrelation accommodated through a first-order autoregressive extension yielding a lagged Chow F -statistic. Candidate partitions are gated by the Bayesian Information Criterion, enabling automatic determination of the number of latent regimes without prior specification. Once stable regimes are identified, regime-specific causal graphs are recovered using DYNOTEARS, and the resulting causal effects are subjected to a formal validation battery comprising refutation tests, sensitivity analysis to unobserved confounding, and graph falsification.

On synthetic accretion data constructed so that the two regimes are distinguishable **only** by causal mechanism, with all variable means equalised across regimes, LIRA recovers the true regime boundary with an Adjusted Rand Index of 0.998, while structure-blind clustering baselines achieve $\text{ARI} \leq 0.012$. Structural coefficients are recovered to within 0.02 of their generating values in both regimes, including the correct identification of the causally isolated variable in each. A single pooled model fitted without segmentation recovers a spurious averaged structure matching neither regime. All estimated causal effects survive a refutation battery, and sensitivity analysis establishes that effect robustness to unobserved confounding scales with effect magnitude ($r = 0.89$), implying that the soft state's weaker radio-X-ray coupling warrants correspondingly weaker causal claims.

Keywords: Causal Artificial Intelligence, Causal Inference, Latent Regime Discovery, Black Hole Accretion, Structural Break Detection, Causal Discovery, DYNOTEARS, Structural Causal Models, Time Series Analysis, Observational Astronomy

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Chapter 1

Introduction

This section introduces the research context, motivation, and objectives of the study. It presents scientific challenges in the search for hidden causal regimes in observational data from black hole accretion and sets the agenda for a causal framework for artificial intelligence that can meet those challenges. The chapter further presents the proposed research objectives, the method used to conduct the research, and the scope and limitations of the research.

1.1 Background

For the past two decades observational astronomy has advanced greatly, both in the quantity and quality of the astrophysical data. The high-cadence, multi-wavelength surveys from missions like *Chandra*, *XMM-Newton*, and *NICER* [7], [9], [18] have turned our view of the universe from a series of static snapshots into a data-rich cinema. Challenge our traditional paradigms for data interpretation to be able to decipher the complexity of the signals we get. Black hole accretion systems are among the most studied astrophysical systems and provide interesting environments to probe high energy physical processes. Black holes are regions of spacetime warped so extremely that the speed needed to escape it has to be greater than the speed of light. We know that speed of light in vacuum is the fastest and is the universal speed limit. Nothing with mass can escape the event horizon and can travel at or faster than this, because it would require infinite energy. However, the regions surrounding these voids are far from silent. Accretion - the process of the gravitational capture and inward spiralling of matter - converts gravitational potential energy into radiation with unparallel efficiency [2], [8]. It is one of the most pervasive and important phenomena in the Universe. Rather than a steady glow, black hole binaries and Active Galactic Nuclei (AGN) shows profound variability across multiple timescales [13], [15]. This variability is not random noise. It manifests as distinct "accretion states" - regimes where the geometry of the flow and the mechanism of energy release undergo fundamental shifts. Active black hole accretion disk transitions occurs primarily across two primary observational regimes - *Hard State* and *Soft State* [11]. Understanding the transitions between these states is critical to decipher the life cycle of a black hole. Despite the obvious complexity of these state transitions, a state of order emerges in the form of power-law scaling relations. Whether observing a stellar mass black hole at our own galaxy or a supermassive black hole at the centre of a distant quasar, certain physical properties appear to

follow universal scaling laws. These mathematical symmetries suggest that the underlying physics of accretion is self-similar across orders of magnitude, hinting at a unified "Fundamental Plane" of black hole activity [10]. Moving beyond mere observation, we must bridge the gap between correlation and causation. In complex astrophysical systems, simply identifying two variables (such as X-ray flux and radio emission) move together is not sufficient enough. We can use *Causal Discovery* to identify direct, functional relationships between physical parameters. This offers a rigorous approach to determine which processes drive others. In the case of a thesis, this represents a shift from *describing* the universe to *diagnosing* its governing mechanisms. However, traditional methods often fall short. For instance, linear correlation models frequently treat accretion systems as "black boxes", ignoring the non-linear dynamics and the time-delayed feedback loops inherent in turbulent plasma. Furthermore, many existing models are forced to assume a specific geometry *a priori*, which may bias the results or overlook *hybrid states* that do not fit into predefined categories. The primary challenge, therefore, lies in the "latent" nature of the system. The true drivers of accretion—the magnetic field configurations, the internal viscosity, and the local shocks—are often not directly observable. We see only the "emergent" light. This thesis addresses the problem of **Latent Regime Discovery**: the task of uncovering the hidden physical states that govern a system's behaviour directly from the observational data. By applying causal inference to the latent variables of accretion disks, we aim to move toward an automated, objective understanding of how black holes consume matter and shape the cosmos.

1.2 Rationale of the Study

Black hole accretion systems exhibit profound variability across multiple timescales, transitioning between physically distinct states. The mechanisms governing jet formation, spectral emission, and energy dissipation differ fundamentally. Even after this, observational datasets arrive unlabelled - the physical state of each observation must be inferred rather than read directly. Current practice relies on manual classification by domain experts, a process that is time-intensive. It also does not scale to the data volumes produced by modern high-cadence instruments such as *NICER*. Beyond the labelling problem, existing causal analyses of accretion systems are limited by their inability to account for regime-dependent changes in causal structure. If we fit a single pooled causal model across observations drawn from physically distinct accretion states, it will produce biased coefficient estimates and obscure the true coupling laws governing jet-disk interaction. If the causal relationship between X-ray luminosity and radio emission obeys one functional form in the hard state and a different form in the soft state, this difference cannot be recovered without first identifying which observations belong to which state. The development of an automated, objective framework for latent causal regime discovery would therefore carry direct scientific value. It would enable the systematic causal characterization of newly obtained observational datasets without manual intervention, and could reveal regime-dependent physical coupling laws that correlational studies are structurally unable to detect. This represents a meaningful contribution to the growing application of causal artificial intelligence to observational science.

1.3 Problem Statement

This thesis addresses the absence of methods capable of discovering latent causal regimes from unlabelled observations without requiring pre-specified environment labels, surrogate domain indices, or prior knowledge of the number of hidden states. Existing causal discovery methods rely on requirements unmet by observational astrophysical data. Specifically, ICP (Peters et al., 2016) [20] and IRM (Arjovsky et al., 2019) [25] require environment labels as a prerequisite. CD-NOD (Huang et al., 2020) [29] requires a surrogate variable encoding the changing conditions—typically a time or domain index—which is unavailable when the latent state is itself unknown. Furthermore, CCSL (Chen et al., 2021) requires multiple independent subject datasets, a structure incompatible with the single continuous observational records typical of X-ray binary monitoring.

Additionally, traditional clustering methods group observations by feature similarity rather than causal mechanism stability. They cannot detect regime boundaries where observable distributions overlap while the underlying causal laws differ. Compounding these limitations, high-cadence accretion data is temporally autocorrelated. Standard causal discovery algorithms assume independent and identically distributed (i.i.d.) observations; when this assumption is violated, the underlying statistical tests produce inflated statistics and spurious regime boundary detections. Ultimately, no existing framework addresses this combination of unlabelled data, an unknown regime count, and temporal dependence simultaneously using invariance-based recursive partitioning.

1.4 Objectives

The general objective of this study is to develop and validate a framework capable of discovering hidden causal regimes and their associated causal structures directly from unlabelled, temporally-dependent observational data without requiring environment labels, a surrogate nonstationarity index, or prior knowledge of the number of latent states, and finally apply this framework to black hole accretion observations. The specific objectives are:

1. To formulate accretion observables within a structural causal framework grounded in established scaling physics, so that the model's assumptions are physically motivated rather than adopted as a generic statistical convenience, and so that its parameters admit a direct physical interpretation.
2. To design a statistically principled criterion for detecting when the causal mechanism governing the data has changed, distinguishing genuine shifts in the underlying physical process from variation attributable to noise, addressing the gap identified in Section 1.3, where existing regime-discovery methods either require labelled environments or offer no formal test of significance for a proposed boundary.
3. To ensure this criterion remains statistically valid under the temporal dependence characteristic of high-cadence monitoring data, since standard methods assume independent observations and produce inflated significance and spurious boundaries when that assumption is violated.

4. To determine the number and location of latent regimes automatically, without requiring the analyst to specify how many physical states are present in advance, a requirement which existing frameworks impose but that cannot be satisfied for observational data where the true regime count is precisely what is unknown.
5. To guard against overfitting the discovered partition, so that the method does not fragment the data into spurious, excessively fine-grained segments in pursuit of a marginally better statistical fit.
6. To recover a distinct, directed causal structure within each discovered regime, enabling direct comparison of how the coupling between physical variables differs across states. This is the central scientific question motivating this work, and one that a single model fitted across all states cannot answer.
7. To validate the proposed framework against data with known ground-truth structure before its application to real observations, so that its regime-recovery accuracy and causal-graph fidelity can be established independently of any uncertainty in the astrophysical interpretation of real accretion data.

1.5 Methodology in Brief

The proposed methodology operates as a two-stage pipeline. The input are a set of unlabelled multi-wavelength accretion observations. All variables are log-transformed, motivated by the power-law structure of black hole accretion physics which renders multiplicative relationships linear in log space and justifies the use of ordinary least squares regression. Stage 1 performs latent regime discovery. A global OLS model is fitted to all data, and a Chow F-test [1] assesses whether a single causal mechanism is sufficient. For time-series data, a first-order autoregressive lag term is added to the design matrix to absorb temporal autocorrelation, yielding a lagged Chow F-statistic with $q = 6$ parameters per model. If invariance is rejected, a systematic search over all candidate splitting variables and thresholds selects the split maximizing the F-statistic. The Bayesian Information Criterion (BIC) [3] then determines whether the split is accepted or the node is declared as a leaf. This process recurses until all partitions satisfy the BIC stopping criterion, with the number of regimes K determined automatically. State 2 recovers a directed causal graph within each stable partition using the DYNOTEARS algorithm, a dynamic extension of the NOTEARS algorithm explicitly designed for time-series data. The framework is validated on synthetic data generated from structural causal models with known graph structures, allowing ground-truth evaluation via the Adjusted Rand Index [4]. Real observational data from the WATCHDOG catalogue [21] and the Merloni et al. (2003) [10] Fundamental Plane compilation are soon to be tested. The overall methodology is summarised in Figure 1.1.

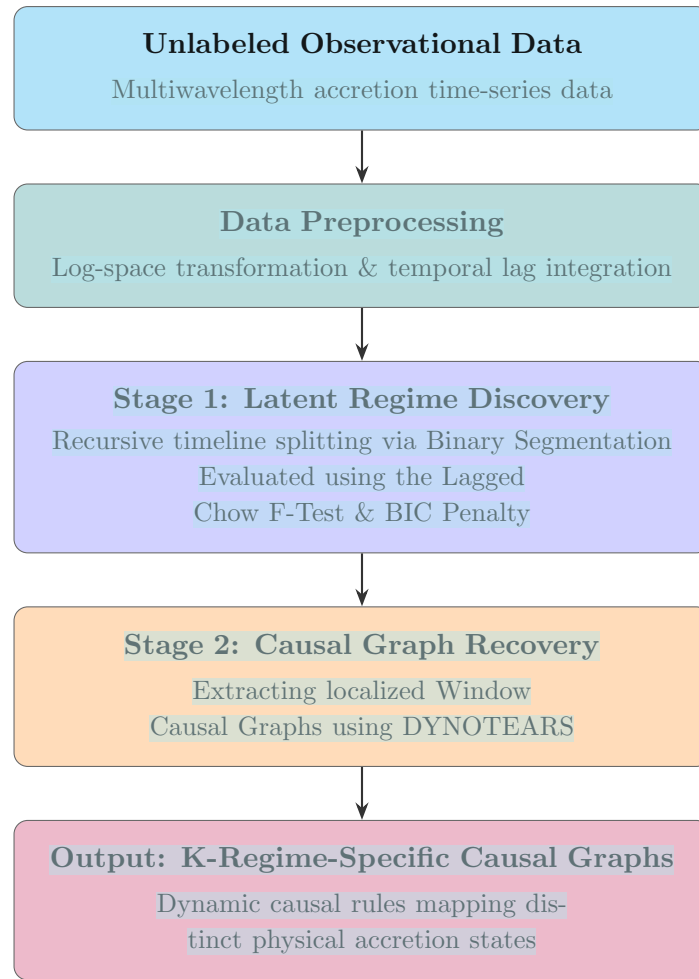


Figure 1.1: High-level pipeline of the unsupervised dynamic causal discovery framework, illustrating the progression from observational data to regime-specific causal graphs.

1.6 Scopes and Challenges

This study lies at the intersection of causal inference and high-energy observational astrophysics. The framework is designed for multi-wavelength black hole accretion data and assumes log-linear structural equations within each discovered regime, consistent with the power-law physics of accretion. It accommodates both feature-based and temporal splitting strategies and applies to any flat observational dataset without requiring multiple independent subject datasets. The output comprises regime-specific causal graphs, regression coefficients, and autoregressive parameters. Several limitations are acknowledged. The linearity assumption within each partition means the genuinely nonlinear intra-regime causal relationships may not be fully recovered at this stage. The piecewise stationarity assumption approximates what are, in reality, continuous accretion dynamics. Gaussian noise is assumed for the derivation of the Chow F-statistic and BIC, which may not hold in all observational settings. The autoregressive model is restricted to lag order one, and empirical validation at this stage relies on synthetic rather than real observational data. Key challenges include hidden confounding from unmeasured physical variables such as magnetic field geometry, which may introduce spurious causal associations that the

PC algorithm [6] cannot resolve without additional background knowledge. Residual temporal autocorrelation not fully absorbed by the lag term may continue to inflate the F-statistic; Newey-West standard errors [5] are considered as a robustness measure. BIC calibration under non-Gaussian observational noise, and the faithfulness assumption required by the PC algorithm [6], present further challenges that will be addressed during validation.

Chapter 2

Literature Review

This chapter provides the theoretical background required to understand the methodology and findings of this thesis, covering the two core fields the work draws from: time-series causal discovery and black hole accretion physics. It begins with a preliminary section that introduces the foundational concepts from both fields, followed by a structured review of the existing literature on causal discovery methods, regime detection, and unsupervised clustering, and concludes with a summary of key findings, observed trends, and the research gaps that this work addresses.

2.1 Preliminaries

Understanding the analyses presented in this thesis requires familiarity with two intersecting areas of knowledge. On one side, a working understanding of how matter accretes onto a black hole and why that process produces fundamentally different observational signatures in the hard and soft states is essential for interpreting what the data actually represents. Black hole accretion systems produce high-cadence, multi-wavelength observational data that reflect transitions between physically distinct states, each governed by its own causal structure linking variables such as X-ray luminosity, radio luminosity, photon index, hardness ratio, and inner disk temperature. On the other side, standard statistical tools are not sufficient to analyse these systems. Accretion light curves are temporally autocorrelated, the regime boundaries are hidden within the unlabelled observations, and the number of distinct physical states is unknown in advance. The tools of causal inference provide a principled framework for going beyond correlation, asking not just which variables move together, but which ones drive which, and whether that driving relationship changes across the latent states of the system. This section introduces the background concepts that underpin the proposed framework, beginning with multivariate time-series analysis and stationarity, moving through the fundamentals of causal inference and structural causal models, and concluding with the algorithmic components of the regime detection and causal discovery pipeline, namely Ordinary Least Squares estimation, the Chow test for structural break detection, Binary Segmentation, the Bayesian Information Criterion, and the DYNOTEARS algorithm.

2.1.1 Multivariate Time-Series and Non-Stationarity

i. Time Series Data

Time series data are a sequence of observations collected at different points in time, where each observation is associated with a specific timestamp. The order of the data, or temporal ordering, is meaningful, and changing this order changes the structure and meaning of the data. More formally, a multivariate time series dataset is denoted as $X = \{x_1, x_2, x_3, \dots, x_n\}$, consisting of n distinct variables where each variable is sequentially measured across T timesteps [36]. In time series data, a specific variable at a given timestep t may be influenced by any other variable in the same timestep, or by variables from previous timesteps ranging from 0 to $t - 1$. The latter is known as lagged effects. Theoretically, a lagged effect can propagate from an infinite earlier point in time. However, assuming an infinite window is not feasible. In a practical sense for Directed Acyclic Graph (DAG) learning, this window is restricted to a designated maximum time lag, represented as I_{max} .

ii. Stationary and Non-Stationary Data

In stationary data, the statistical properties of the data like mean and variance remain constant along every value of T . The data may fluctuate here and there, but the statistical properties never truly shift along any time. On the other hand, in non-stationary data, the statistical properties may shift systematically or totally unpredictably at any time.

iii. Temporal Regime

In time series analysis and causal discovery, a temporal regime is defined by a consecutive window of time in which the data's statistical properties, rules, or causal structure remains constant. For example, let's assume for a time-series dataset, timestep $t = 1$ to $t = m$ has the same properties. So, $t = 1$ to m belong in the same regime. But the properties shift from $t = m + 1$ and stay constant until $t = k$. So, $t = m + 1$ to k belongs to another regime.

iv. Strict Stationary Assumption

Many traditional time series and causal discovery models heavily rely on a strict stationary assumption, meaning they assume that the underlying rules and patterns of the data remain constant along every timestep. In this assumption, the time index is considered to be independent of the data distribution, which implies that the causal dependencies remain unchanged over the whole period of time. However, in dynamic real-world systems, this assumption fails because most real-life data is non-stationary and its underlying distribution shifts following a pattern or randomly to a different temporal regime. Traditional models cannot address this problem and may fail to produce accurate discoveries because they cannot address the changing regimes or multi-regime properties.

2.1.2 Black Holes and Accretion

i. Black Holes

Black holes are among the most mysterious and most studied phenomena in the Universe. Much progress has been made in our knowledge and understanding of these systems, but many aspects of their behaviour are the subject of intensive investigation and are not yet fully understood.

A black hole is a region of space-time that is so dense that nothing, not even light, can escape once it crosses a boundary called the event horizon. Black holes are not directly observable via emitted light but are inferred through their gravitational and electromagnetic effects on nearby matter. Black holes can form when a massive star dies. Einstein's General Theory of Relativity tells us that mass and energy warp the fabric of space-time. The most extreme examples of this curvature are black holes. The event horizon is the boundary beyond which information cannot return to an external observer.

For a non-rotating black hole, the radius of the event horizon is described as the *Schwarzschild Radius*:

$$R_s = \frac{2GM}{c^2} \quad (2.1)$$

where G is the gravitational constant, M is the mass of the black hole, and c is the speed of light. At this radius, the escape velocity is greater than the speed of light, which means that escape is physically impossible according to our current understanding of the universe. Black holes themselves do not emit any electromagnetic radiation, but the matter around them can become extraordinarily luminous through *accretion*.

ii. Accretion and Accretion Disks

Accretion is the primary mechanism by which black holes grow and become observable. It is the process of gravitational capture and the inward spiralling of matter toward a central massive body. In the vacuum of space, most matter has some degree of *spin* or angular momentum. Gas and dust do not fall directly into a black hole in a straight line. Instead, they swirl around it. During this process, the gravitational potential energy of the falling matter is converted into kinetic energy and heat. This conversion is highly efficient, with accretion onto black holes representing one of the most efficient known mechanisms of energy release [2], [8]. As a result, the surrounding regions can become among the most luminous astrophysical sources in the Universe.

As matter approaches a black hole, it does not fall straight in, but instead forms a large, thin disk, spiralling around the black hole. This disk is called an accretion disk. Inside the disk, internal friction (viscosity) causes the gas to lose its orbital energy and slowly spiral inwards. The friction generates enormous heat, with temperatures in the millions of degrees.

The disk's temperature is not uniform; the outer regions are relatively cooler, while the innermost regions reach such extreme temperatures that they glow brightly in the X-ray part of the electromagnetic spectrum. This emission allows astronomers to map the geometry and behaviour of the disk even though the black hole itself remains dark.

iii. Black Hole X-ray Binaries

Black holes are generally divided into three categories according to their mass: *stellar-mass*, *supermassive*, and *intermediate-mass*. The mass ranges that define each group are approximate. In our own galaxy, we often find *stellar-mass* black holes (typically 5 to 30 times the mass of our Sun) in binary systems. These are known as Black Hole X-ray Binaries (BHXBs).

In these systems, the black hole orbits a *normal* companion star. The black hole's gravity is strong enough to strip gas from the outer layers of the companion star. This captured gas feeds the accretion disk, creating a laboratory where we observe rapid changes in the black hole's *feeding habits* over timescales of weeks or months. These are known as Black Hole X-ray Binaries (BHXBs) [13], [15].

iv. Active Galactic Nuclei (AGN)

On a much larger scale, *Active Galactic Nuclei (AGN)* represent supermassive black holes at the centres of distant galaxies. These black holes are millions or even billions of times more massive than our Sun.

Despite the vast difference in size, the underlying physics of how they consume matter remains the same as their smaller *stellar-mass* cousins. Because they are so large, the changes in their behaviour happen much more slowly. These changes take thousands or millions of years, which is why we use smaller X-ray binaries as *fast-forward* models to understand the life cycles of these galactic giants.

v. Accretion States

Black holes do not consume matter at a steady or unchanging rate. Instead, they transition between distinct *accretion states* - physical regimes where the geometry of the gas flow and the mechanism of energy release change fundamentally. These states are identified by the spectrum of the light they emit.

vi. Hard State

In the *Hard State*, the black hole is typically at a lower accretion rate. The accretion disk is often *truncated*, meaning the bright, inner part of the disk is replaced by a hot, diffuse cloud of gas called a *corona*.

A defining feature of this state is the presence of a steady *radio jet* (a powerful beam of particles shot out from the centre at nearly the speed of light). The light we observe is dominated by high-energy (or *Hard*) X-rays, which flicker rapidly.

vii. Soft State

When the black hole begins to consume matter more rapidly, it transitions into the *Soft State*. In this regime, the accretion disk moves all the way inward to the closest possible stable orbit.

In the Soft State, the radio jet is *quenched* or disappears entirely. The emission is dominated by the steady heat of the disk itself, appearing as a *Soft* X-ray glow. This state is disk-dominated, whereas the Hard State is jet-dominated.

viii. Power-Law Scaling Relations

In contrast to the intricate physics of plasma dynamics and magnetic fields, the systems surrounding black holes exhibit a notable level of order. They adhere to Power-Law Scaling Relations. One physical feature (like brightness) is proportional to another raised to a specific exponent, $y = Ax^k$.

In science, power laws are often a sign of scale *invariance*. This suggests that the same physical law applies whether the black hole is small or supermassive. For the purpose of this research, these relations are very important because power laws become linear when converted to a logarithmic scale ($\log y = k \log x + \log A$). This makes them ideal for the causal discovery methods explored here.

ix. Fundamental Plane of Black Hole Activity

The *Fundamental Plane of Black Hole Activity* is a specific scaling relation that serves as a "universal rulebook" for black hole physics [10]. It connects three core properties: Radio luminosity (L_R), X-ray luminosity (L_X), and the mass of the black hole (M_{BH}). Empirical observations indicate that the physical processes involved in accretion and jet production are similar over several orders of magnitude in mass:

$$L_R \propto L_X^{\xi_X} M_{BH}^{\xi_M} \quad (2.2)$$

This relation is expressed logarithmically as:

$$\log L_R = a \log L_X + b \log M_{BH} + c \quad (2.3)$$

2.1.3 Fundamentals of Causal Inference

To understand the mechanisms governing each temporal regime, we rely on the formal framework of causal inference. This section introduces the necessary graphical and structural definitions, adopting the framework detailed by Neal (2020).

i. Directed Acyclic Graphs (DAGs)

A *graph* is a mathematical structure consisting of a set of *nodes* (or vertices) and a set of *edges*, where each edge represents a relationship between two nodes. In the context of causal inference, nodes represent the variables or features of the dataset [32].

In a *directed graph*, edges contain arrows indicating the direction of the relationship. The node from which an edge originates is called the *parent* node, and the node the edge points to is called the *child* node.

A *path* is a sequence of nodes connected by edges, regardless of the direction of the arrows. A *directed path*, however, is a sequence of nodes connected by directed edges that all point in the same forward direction.

For a given node X , its *descendants*, denoted as $de(X)$, consist of all nodes Y for which there exists a directed path from X to Y (this includes children, children of children, and so forth). Conversely, the *ancestors* of Y , denoted as $an(Y)$, consist of all nodes X for which there exists a directed path from X to Y (including parents, parents of parents, etc.).

A **Directed Acyclic Graph (DAG)** is a directed graph that contains no *cycles*; namely, there does not exist any directed path that starts and ends at the same node. Consequently, no node can be an ancestor or descendant of itself [32].

ii. Causal Graphs

A variable X is considered a *cause* of a variable Y if an intervention or deliberate change in X results in a change in the distribution of Y ; this underlying mechanism is referred to as *causation* [32].

Within the framework of a directed graph, the *Causal Edges Assumption* states that every parent node is a direct cause of all its child nodes.

Consequently, a **Causal Graph** is formally defined as a Directed Acyclic Graph (DAG) that strictly satisfies the Causal Edges Assumption. In such a graph, every directed path from one node to another implies a flow of causation.

iii. Statistical Independence and Dependence

Two random variables X and Y are **statistically independent** (or marginally independent), denoted as $X \perp\!\!\!\perp Y$, if for all possible realized values x and y , their joint probability satisfies:

$$P(X = x, Y = y) = P(X = x)P(Y = y) \quad (2.4)$$

If this equality does not hold for at least one pair of (x, y) , the variables are **statistically dependent**, denoted as $X \not\perp\!\!\!\perp Y$, implying that observing the value of X provides information that changes the probability of Y .

In real-life systems, variables are rarely entirely independent. Instead, we rely on **conditional independence**. Two random variables X and Y are conditionally independent given a third variable (or set of variables) Z , denoted as $X \perp\!\!\!\perp Y \mid Z$, if for all possible realized values x, y , and z :

$$P(X = x, Y = y \mid Z = z) = P(X = x \mid Z = z)P(Y = y \mid Z = z) \quad (2.5)$$

In causal terms, conditional independence implies that once Z is known, observing X provides no additional information about Y [12].

iv. d-Separation and the Flow of Association

In a causal graph, it is necessary to understand the conditional independence or dependence between two nodes. This criterion is known as **d-separation** (directional separation) [32].

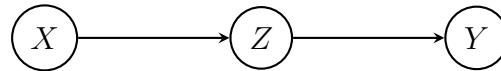
Two nodes X and Y are d-separated by a set of conditioning nodes Z if all paths between X and Y are mathematically blocked by Z . If X and Y are d-separated in the graph $\mathcal{G}$, it implies they are conditionally independent in the probability distribution: $X \perp\!\!\!\perp Y \mid Z$. Note that Z is a set of nodes which can be empty.

To determine whether a specific path is open or blocked, we evaluate the path using three structural building blocks [32]:

1. Chains (Mediation)

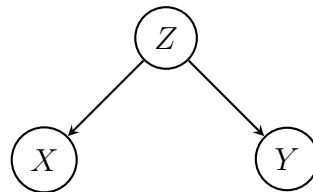
A chain represents a sequence where causation flows through an intermediate variable ($X \rightarrow Z \rightarrow Y$). Here, X is a cause of Z , and Z is

a cause of Y . This path is by default open and X and Y are statistically dependent. However, conditioning on the mediator Z blocks the flow of association. *Example:* In urban infrastructure, heavy rain (X) causes waterlogging (Z), which subsequently causes severe traffic gridlock (Y). Conditioning on the waterlogging (Z) mathematically blocks the association between heavy rain and traffic gridlock.



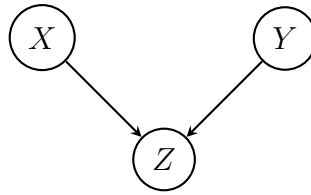
2. Forks (Confounding)

A fork happens when a single variable is the common cause of two separate variables ($X \leftarrow Z \rightarrow Y$). Here, Z is a cause of both X and Y . This path is open by default, introducing confounding bias. Conditioning on the common cause Z blocks this spurious flow of association. *Example:* A summer heatwave (Z) simultaneously causes load-shedding (X) and an increase in app-based ride demand (Y). Conditioning on the temperature (Z) d-separates the electricity usage from the ride-hailing demand, removing the fake correlation between load-shedding (X) and app-based ride demand (Y).



3. Colliders (Selection Bias)

A collider exists when two independent variables have a common effect ($X \rightarrow Z \leftarrow Y$). Here, both X and Y are causes of Z . If X and Y are not directly connected by an edge, this specific V-structure is formally known as an *immorality*. Unlike chains and forks, a collider naturally blocks the flow of association. However, conditioning on the collider Z (or any of its descendants) opens a spurious path between X and Y . *Example:* In an academic environment, consider the factors for getting a good grade in a certain course (Z). A student's prior foundational knowledge (X) and their dedicated study hours (Y) independently increase the probability of securing an 'A'. Prior knowledge and study hours might be independent; however, if a researcher restricts their analysis to students who actually received an 'A' (conditioning on the collider Z), they will induce a spurious negative correlation. Within this conditioned group, a student without prior foundational knowledge must have compensated with a high number of study hours to achieve the grade. This creates a mathematically false implication that having stronger foundational knowledge causes a student to study less.



Formally, a path between nodes A and B is blocked (no flow of association) by a conditioning set M if at least one of the following conditions is met [32]:

1. The path contains a chain ($X \rightarrow Y \rightarrow Z$) or a fork ($X \leftarrow Y \rightarrow Z$) such that the middle node Y is in the conditioning set M .
2. The path contains a collider ($X \rightarrow Y \leftarrow Z$) such that the middle node Y is *not* in M , and no descendant of Y is in M .

v. Structural Causal Models (SCMs)

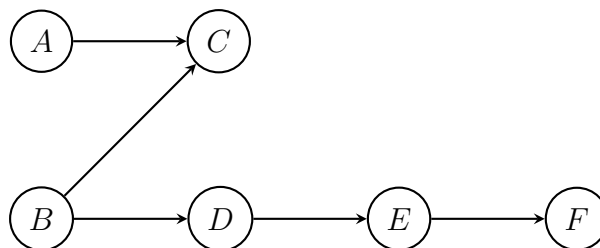
A **Structural Causal Model (SCM)** is a mathematical framework used to model cause-and-effect relations in an environment [32]. While a DAG is the visual description of causality, an SCM provides the mathematical rules for those relations. In an SCM, every single node is calculated through a function:

$$X_i := f_i(\text{PA}(X_i), \epsilon_i) \quad (2.6)$$

where:

- X_i (The effect): The variable being calculated.
- $\text{PA}(X_i)$ (The causes): The set of parents of X_i which are causing the effect.
- f_i (The function): The deterministic mathematical relation that computes the effect from the causes.
- ϵ_i (The noise term): The variation in the effect that the causes *cannot* explain.
- $:=$ (The assignment): The mathematical assignment operation means X_i is defined by the right-hand side. **Note that this operation is not symmetric.**

To illustrate how a DAG translates into structural mathematical equations, consider a system with the causal structure: $A \rightarrow C$, $B \rightarrow C$, $B \rightarrow D$, $D \rightarrow E$, and $E \rightarrow F$.



The corresponding SCM defines each node as a deterministic function of its direct parents and its unobserved noise (ϵ):

$$\begin{aligned} A &:= f_A(\epsilon_A) \\ B &:= f_B(\epsilon_B) \\ C &:= f_C(A, B, \epsilon_C) \\ D &:= f_D(B, \epsilon_D) \\ E &:= f_E(D, \epsilon_E) \\ F &:= f_F(E, \epsilon_F) \end{aligned}$$

In this structure, nodes A and B have no direct parents (they act as root nodes), meaning their assignments rely entirely on their respective noise terms. Node C , acting as a collider between A and B , takes both parent variables as arguments in its structural function.

vi. The Discovery Postulates: Markov and Faithfulness

Every mathematical field stands on axioms and postulates. Similarly, to successfully discover a causal graph $\mathcal{G}$ from the probability distribution P of observational data, it is necessary to establish a mathematical bridge between the SCM and statistical independence. This bridge relies on two foundational postulates: the **Causal Markov Assumption** and the **Faithfulness Assumption** [32].

1. The Local Causal Markov Assumption

The Causal Markov Assumption dictates that a node is conditionally independent of all its non-descendants, given its direct parents. Formally, for any node X in a DAG $\mathcal{G}$, where $\text{PA}(X)$ represents its parents and $\text{ND}(X)$ represents its non-descendants:

$$X \perp\!\!\!\perp \text{ND}(X) \mid \text{PA}(X) \quad (2.7)$$

This assumption guarantees that the causal graph naturally imposes the correct conditional independencies on the joint probability distribution. If two variables are d-separated in the graph $\mathcal{G}$, they *must* be statistically independent in the data P .

2. The Faithfulness Assumption

The Faithfulness Assumption is the logical converse of the Markov Assumption. A probability distribution P is considered faithful to a DAG $\mathcal{G}$ if *every* conditional independence observed in the data is strictly the result of a structural d-separation in the graph. Formally, if $X \perp\!\!\!\perp Y \mid Z$ is found in the data distribution P , then X and Y must be d-separated by Z in $\mathcal{G}$. This axiom ensures that two causal paths do not perfectly and coincidentally cancel each other out to create a fake statistical independence.

Together, these postulates guarantee that the structural d-separations in the causal graph perfectly map to the statistical conditional independencies in the observable time-series data, making the unsupervised discovery of the DAG mathematically possible.

2.1.4 Unsupervised Learning and Clustering

Because the structural boundaries of real-world temporal regimes are rarely known in advance, discovering these shifts requires analyzing the data without predefined labels. This section discusses the application of clustering and temporal segmentation to identify these hidden causal boundaries.

i. The Need for Unsupervised Learning

We have theoretically established the basic theoretical framework for modeling causal structure. In this section, we address a major methodological challenge of causal modeling, which is detecting the temporal regimes without knowing the structural boundaries of the data. We frame this as an unsupervised learning problem. In traditional supervised learning algorithms, models are trained with data with input features x where the actual outputs y are known beforehand. But in most real life dynamic systems, it is not feasible to gather the predefined target outputs. So, the exact boundaries of the temporal regimes stay hidden. Supervised learning cannot work in this case. To solve this, we rely on unsupervised learning where we try to find the hidden structures of the data without referring to any known or labeled outcomes [16]. Since our multivariate time-series data does not contain the predefined labels, it is mandatory to use unsupervised learning to discover the underlying Structural Causal Model (SCM) shifts.

ii. Clustering vs. Temporal Segmentation

One of the primary techniques for unsupervised learning is the method of clustering. In this method, the data are grouped in a way such that the intra-group similarity is maximized and the inter-group similarity is minimized. One such well known clustering technique is k-means clustering. In this method, data points are assigned to clusters based entirely on their feature proximity, ignoring the sequence of the data. So, applying standard clustering on time-series data where the sequence matters, destroys the temporal precedence required for causal discovery. To solve this problem, we can use temporal segmentation. Temporal segmentation safeguards continuity by grouping contiguous time blocks into discrete clusters. Here, each cluster represents a temporal regime, and the boundary between two regimes is now clearly defined, which our causal model may work with [36].

To illustrate this within a real-world context, consider the highly dynamic environmental and traffic conditions of Dhaka City. A multivariate time-series dataset tracking hourly parameters, such as traffic density, weather metrics (humidity, wind speed), and air quality markers like PM2.5 exhibits distinct, unlabeled temporal regimes. During the Winter Regime (November-February), cold weather and stagnant air mean traffic density has a strong, direct causal link to sudden PM2.5 spikes. However, during the Monsoon Regime (June-August), heavy rainfall washes away pollutants, breaking or heavily weakening that direct causal link, while the rain itself becomes the primary cause of traffic gridlock. Standard clustering like k-means would completely ignore this and may group data based only on features like traffic speed, which may cluster winter data and summer data together. Temporal segmentation, however, treats winter and monsoon as continuous, separate blocks of time with their own distinct causal behaviors.

iii. Requirements for Temporal Segmentation

Although there exist many algorithms which may be used to perform temporal segmentation, some rules or requirements have to be enforced for our specific problem.

- **Contiguity enforcement:** The algorithm must ensure that each cluster is an unbroken sequence of time.
- **Dynamic graph optimization:** The cluster fit shouldn't just be the distance between two data points, but also by how a Directed Acyclic Graph (DAG) explains the entire block of data.
- **Complexity penalization:** The algorithm might assign each time-step its own unique regime which we have to prevent. We can prevent it by ensuring that some penalty term is added to the objective function which balances the accuracy of the causal graph against the number of structural breaks.

By satisfying the requirements, the unsupervised algorithm can successfully divide the non-stationary time-series data into unique stationary regimes, which will allow the discovery of a DAG within each time block.

2.1.5 Dynamic Causal Models and Time-Series Graphs

In a standard Structural Causal Model (SCM), cause-and-effect relationships are captured only at a single, isolated instant. The underlying mathematics cannot account for historical dependencies, and a traditional DAG is limited to representing static relations. To evaluate dynamical systems, causal influences from the past must be explicitly taken into account. Therefore, to properly study structural breaks across temporal regimes, we must shift from the standard causal framework introduced in the previous section to Dynamic Causal Models.

i. Unrolling the Graph over Time (Dynamic Bayesian Networks)

A Dynamic Bayesian Network (DBN) is a directed graphical representation of a stochastic process [19]. It is important to note that the network structure itself is not dynamic; rather, it is a fixed topology that models variables across discrete slices of time. By unrolling these static time slices, the network formally represents a dynamical system capable of accounting for historical dependencies.

To rigorously define a DBN, three structural requirements must be specified [19]:

1. **The Initial State:** The structural topology of the first time-slice ($t = 0$).
2. **The Transition Model:** The directed connections between historical time-slices ($t - \tau$) and the current time-slice (t).
3. **The Probability Distributions:** The specific Conditional Probability Distributions (CPDs) that mathematically govern the relations along these directed connections.

ii. Defining Time-Series Causal Graphs

To apply causal discovery algorithms to time-series data, we must mathematically formalize how the unrolled Dynamic Bayesian Network is represented as a graph. Depending on the level of temporal detail required, time-series causality can be represented using three distinct graphical definitions [36]. These representations are illustrated in Figure 2.1:

1. The Full Time Causal Graph

The Full Time Causal Graph captures all causal relations throughout all possible time. This is an infinite Directed Acyclic Graph (DAG) where every variable at every time step is denoted by an independent node. Formally, the set of nodes is defined as $\{X_t^i \mid \forall i, \forall t\}$. An edge drawn from $X_{t-\tau}^j \rightarrow X_t^i$ indicates a direct causal relationship with an exact time delay of τ . If $\tau = 0$, the edge represents a contemporaneous effect; if $\tau > 0$, the edge represents a lagged effect. Although this graph contains all the theoretical information, the sheer number of nodes and edges makes it computationally intractable for causal discovery algorithms.

2. The Window Causal Graph

Because the Full Time Causal Graph is computationally intractable, we must compress the temporal information into a manageable window. As a result, this graph restricts the analysis to a specific, finite maximum time lag, denoted as τ_{max} . In this representation, an edge $X_{t-\tau}^j \rightarrow X_t^i$ is only valid and included in the graph if the delay is within the sliding window ($\tau \leq \tau_{max}$). This operates under the assumption that an effect past τ_{max} is negligible and is instead mediated by more recent causal relations. The Window Causal Graph is the primary target structure for time-series causal discovery algorithms and is the most prominent representation utilized in this thesis.

3. The Summary Causal Graph

Unlike the Full Time Causal Graph or the Window Causal Graph, the Summary Causal Graph does not incorporate timestamps; it serves as a full summary of the entire temporal history, containing information across all time steps. In this graph, nodes revert to their static definitions (X^i and X^j , dropping the time index t). An edge $X^j \rightarrow X^i$ is drawn if there exists *any* time delay $\tau \geq 0$ where $X_{t-\tau}^j$ causes X_t^i in the underlying full graph. Because this aggregation ignores the strict forward-flowing direction of time, a Summary Causal Graph can contain cycles ($X^j \rightarrow X^i$ and $X^i \rightarrow X^j$ can simultaneously exist if they occur at different lags). Therefore, unlike a standard static causal graph, a Summary Causal Graph is not necessarily a DAG.

By understanding these three representations, we can define unsupervised clustering algorithms to discover distinct temporal regimes in a time-series by analyzing localized Window or Summary Causal Graphs.

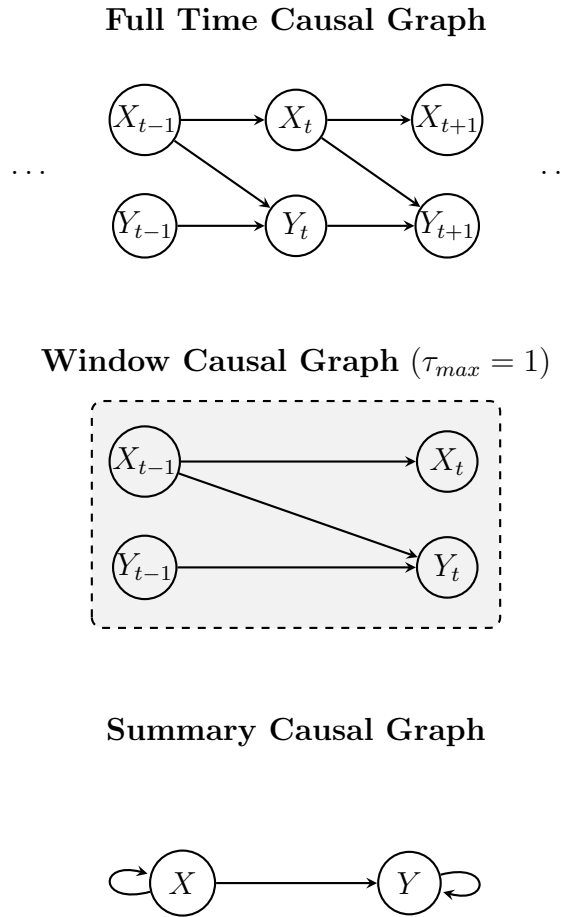


Figure 2.1: Representations of Time-Series Causal Graphs

iii. The Time-Series Structural Causal Model

A standard Structural Causal Model (SCM) assumes all variables operate within the same isolated instant. However, in a Time-Series Structural Causal Model, we must extend this framework to account for the flow of time t .

In a time-series SCM, a variable is not merely caused by other variables, but by specific historical time instants (from $t - 1$ down to $t - \tau_{max}$) of those variables [36]. We formally define a time-series SCM as:

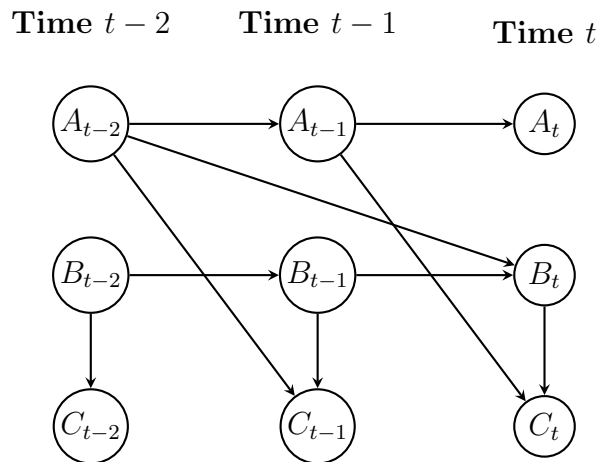
$$X_t^i := f_i(\{X_{t-\tau}^j \mid X_{t-\tau}^j \in \text{PA}(X_t^i)\}, \epsilon_t^i) \quad (2.8)$$

where:

- X_t^i (The present effect): The target variable being calculated at the current time step t .
- $\text{PA}(X_t^i)$ (The temporal parents): The set of all direct causes of X_t^i . Because time is now a factor, this set can contain two distinct types of causes:
 - **Contemporaneous Effects:** Causes that occur within the exact same time step ($\tau = 0$), denoted as $X_t^j \rightarrow X_t^i$.
 - **Time-Lagged Effects:** Causes that occurred in the strict past ($\tau > 0$), denoted as $X_{t-\tau}^j \rightarrow X_t^i$. Note that a variable can be a lagged cause of itself ($X_{t-\tau}^i \rightarrow X_t^i$), representing auto-correlation.

- f_i : The deterministic function mapping the historical and instantaneous parents to the current effect.
- ϵ_t^i : The independent noise term at time t .

To illustrate how a static DAG is unrolled into this time-series SCM format (using a maximum time lag of $\tau_{max} = 2$), consider a dynamic system with three variables (A , B , and C). Let the causal structure dictate the following temporal rules: A influences its own next state, A influences B with a delay of two time steps, B influences its own next state, A influences C with a delay of one time step, and B influences C instantaneously.



The corresponding time-series equations dictate the state of each node at the current time step t :

$$\begin{aligned} A_t &:= f_A(A_{t-1}, \epsilon_{A,t}) \\ B_t &:= f_B(B_{t-1}, A_{t-2}, \epsilon_{B,t}) \\ C_t &:= f_C(B_t, A_{t-1}, \epsilon_{C,t}) \end{aligned}$$

In this dynamic system, the equation for A_t demonstrates a lagged auto-correlation ($\tau = 1$), the equation for B_t demonstrates a cross-lagged effect reaching back to the maximum lag limit ($\tau = 2$), and the equation for C_t demonstrates a combination of a historical effect from A ($\tau = 1$) and a contemporaneous effect from B ($\tau = 0$).

iv. Consistency Throughout Time and Structural Breaks

The mathematical formalizations of Time-Series SCMs and Window Causal Graphs rely on a fundamental assumption: the underlying causal mechanisms do not change spontaneously or erratically. It is highly unlikely that a causal direction will abruptly flip within a stable window; if such instability occurred frequently, the system would fragment into countless micro-regimes, causing the predictive model to collapse. In most macroscopic real-world scenarios, a causal link from A to B might weaken or disappear over time, but an immediate reversal where B suddenly causes A is highly improbable. This principle of stability is formally known as **Consistency Throughout Time**.

According to Assaad et al. [36], a time-series causal graph is consistent throughout time if the presence and direction of every causal edge remain invariant across all time steps. For instance, if X causes Y with a lag of $\tau = 1$ at time t , it must also cause Y with the exact same lag at time $t + 100$.

In the context of this thesis, Consistency Throughout Time serves as the core defining characteristic of a **Temporal Regime**. As long as the consistency of the causal graph remains unbroken, that contiguous block of time constitutes a single, independent regime.

However, in real-world dynamical systems, environments evolve and underlying rules change. When this consistency assumption is violated over time—for example, when an existing causal edge disappears or a new one emerges—we denote that exact moment as a **Structural Break**. Consequently, the time steps following this break transition into an entirely different regime. In observational data, this often manifests as certain variables becoming causally isolated while new cause-and-effect relationships are born.

To illustrate this dynamically, we extend the Dhaka City environmental model introduced in Section 2.1.4. Consider a multivariate time-series dataset tracking hourly parameters: Traffic Density (X), PM2.5 levels (Y), and Rainfall (Z).

- **Regime 1: The Winter Consistency (Nov–Feb):** Cold weather and stagnant air mean traffic density has a strong, direct causal link to sudden PM2.5 spikes. During this contiguous block of time, the Window Causal Graph maintains a perfectly consistent, lagged edge: $X_{t-\tau} \rightarrow Y_t$.
- **The Structural Break:** As the climate shifts, the environmental physics governing the city fundamentally change. The consistency assumption is violated.
- **Regime 2: The Monsoon Consistency (Jun–Aug):** Heavy rainfall washes away pollutants, effectively breaking the direct causal link from traffic to air quality. Simultaneously, the rain itself becomes the primary cause of traffic gridlock. The old causal edge ($X_{t-\tau} \rightarrow Y_t$) disappears from the Window Causal Graph, replaced by a newly consistent edge representing the rain’s effect on traffic ($Z_{t-\tau} \rightarrow X_t$).

Because the exact temporal boundary (the structural break) between these distinct causal regimes is hidden within the unlabeled observational data, we cannot rely on supervised learning to locate it. This dictates the core methodological approach of this thesis: we must treat the discovery of these structural breaks as an **unsupervised learning** problem, utilizing clustering algorithms to segment the time-series based solely on the shifting structures of the underlying Window Causal Graphs.

2.1.6 Algorithmic Framework for Regime Detection and Causal Discovery

Identifying independent temporal regimes, and more precisely, locating their underlying structural breaks requires a multi-step pipeline. Therefore, our methodology relies on the integration of multiple algorithms, analytical tools, and optimization techniques. Within this pipeline, the time-series data is first segmented into distinct

temporal regimes, and subsequently, the specific causal structure of each isolated regime is extracted. To achieve this, our framework relies on three core stages:

i. Parameter Estimation and Model Scoring

- **Ordinary Least Squares (OLS):** OLS serves as the baseline optimization technique in this framework. It mathematically estimates the strength of the relationship (the edge weight) between a historical cause and a present effect. Because these cause-and-effect relationships are temporal, the true edge weights remain constant within a stable regime but shift when a structural break occurs. By minimizing the residual sum of squares, OLS evaluates the baseline error for any given window of time.
- **Bayesian Information Criterion (BIC):** BIC acts as the penalized scoring metric for this framework. During clustering, if every tiny chunk of time were segmented independently, the OLS error would drop to near zero, but the model would severely overfit, rendering the results meaningless. BIC introduces a mathematical penalty for each new regime discovered. Consequently, it balances the need to minimize OLS error against model complexity, ensuring the algorithm does not fracture the timeline into an excessive and insignificant number of segments.

This fundamental trade-off between minimizing OLS error and minimizing the BIC penalty ensures the mathematical foundation necessary to deploy search algorithms to locate the exact structural breaks.

ii. Structural Break Identification

To identify the exact temporal boundaries where structural breaks occur, we utilize recursive hypothesis testing driven by OLS error evaluation.

- **Binary Segmentation (BinSeg):** Because a time-series may contain multiple hidden regimes, evaluating every single time step independently for a structural break is computationally intractable. To optimize this, we employ Binary Segmentation, a recursive "divide-and-conquer" search strategy. At each step, the algorithm sweeps the current time window, identifies the single point with the highest Chow Test significance score, and divides the data at that exact juncture. The recursion then continues independently on those newly divided sub-segments. Furthermore, because the true number of regimes is unknown, this method is highly advantageous; it iteratively halves the timeline only when statistically justified, isolating independent regimes without requiring a predefined regime count.
- **Chow F-Test:** The Chow F-Test serves as the statistical engine that dictates whether BinSeg should actually execute a split. It evaluates if a structural break is genuinely happening by comparing the OLS error of a single continuous time window against the combined OLS error of the two segments proposed by BinSeg. If formulating two distinct regression models significantly reduces the overall residual error, the Chow Test mathematically confirms that a structural break has occurred and the regime has shifted.

iii. Continuous Causal Graph Discovery

Once the entire time-series is segmented into independent regimes, the final step is to discover the internal dynamic causal graph governing each regime.

- **The DYNOTEARS Algorithm:** Traditional causal discovery algorithms, such as the PC algorithm, rely on combinatorial graph search and do not scale well with dense time-series data. To extract the causal relations and underlying structures from each independent regime, we utilize DYNOTEARS, a temporal extension of the NOTEARS framework. DYNOTEARS advances DAG learning by replacing discrete graph search with a continuous, differentiable optimization problem. This formulation allows the algorithm to efficiently learn both instantaneous and lagged causal effects simultaneously. By applying DYNOTEARS individually to each temporal block isolated by BinSeg, we can extract the exact localized Window Causal Graph for that specific regime.

iv. Summary and Baseline Assumptions

In summary, this pipeline functions as a comprehensive framework for unsupervised causal inference: OLS measures the baseline error, BIC prevents model overfitting, Binary Segmentation recursively divides the timeline, the Chow F-Test statistically validates the structural breaks, and finally, DYNOTEARS maps the dynamic causal graphs.

It must be noted that this baseline framework initially operates under the assumption of linear causal relationships. If real-world datasets exhibit strong non-linear mechanisms, the performance of this specific configuration may degrade. However, because this pipeline is inherently modular, each algorithm or statistical test can be substituted with non-linear alternatives optimized for the target dataset. Consequently, this architecture can be adapted to capture the behaviors of highly complex dynamic systems.

2.2 Causal Discovery and Time Series: Progress and Key Findings

Over the past two decades, causal discovery has evolved from classical graphical models to flexible, often deep-learning-based methods that handle mixed data, heterogeneity, and temporal dependence. More recent work focuses specifically on time-series causal discovery, including nonstationary and high-dimensional settings.

2.2.1 Graphical Causal Discovery

The mathematical formalization of causal discovery has traditionally bifurcated into constraint-based frameworks, such as PC and Fast Causal Inference (FCI) algorithms, and score-based Directed Acyclic Graph (DAG) learning paradigms. While these classical methodologies established the theoretical bedrock of structure learning, their practical deployment often exposes fundamental vulnerabilities to real-world data pathologies. As reviewed by Glymour et al. [26], issues such as severe non-linearity, measurement error, nonstationary time series, and latent confounding

consistently undermine the asymptotic guarantees of early algorithms. To mitigate the computational bottlenecks of pure score-based searches and the statistical sensitivity of pure constraint-based methods, recent literature has shifted toward hybrid frameworks. For instance, Kuipers et al. [22] demonstrated that constraint-based filtering can be used to drastically prune the search space, enabling efficient Markov Chain Monte Carlo (MCMC) sampling over the remaining DAG space. This integration ultimately facilitates robust Bayesian model averaging even across large-scale networks.

While early structural learning algorithms faced strict data and architectural limitations, recent advances have significantly broadened their scope and reliability. For instance, handling mixed continuous and discrete data has become more robust through the use of likelihood-ratio conditional independence tests [23], while variational Bayesian frameworks have made learning scalable even for large sets of variables [34].

Parallel efforts have focused on making constraint-based algorithms more dependable in practice. Recent work introduces internal incoherency scores to catch finite-sample errors and assumption violations [45], and the theoretical soundness of FCI-type approaches has now been extended to handle cyclic graphs [30]. Furthermore, modern Bayesian methods allow researchers to jointly learn DAG structures and causal effects, which effectively propagates structural uncertainty directly into intervention estimates [35]. Finally, the frontier of mixed-type structure learning has been pushed forward by adapting NOTEARS-style continuous optimization to diverse data domains [43].

2.2.2 Heterogeneity, Non-stationarity, and Mechanism Shifts

Modern work exploits distribution shifts across environments as a signal for causal directions. Huang et al. [29] developed CD-NOD, which learns causal skeletons and directions from heterogeneous or nonstationary data and estimates low-dimensional driving forces of changing mechanisms. A related survey highlights non-stationary time series and heterogeneous multi-set data as central practical challenges [26]. The Sparse Mechanism Shift hypothesis proposes that only a few conditionals change across environments; the Mechanism Shift Score (MSS) can then identify the full causal structure nonparametrically under such sparse shifts [37].

New methods explicitly cluster heterogeneous causal mechanisms: Li et al. [46] jointly infer latent clusters and their causal graphs from mixed observational data without environment labels. At the bivariate level, Liu et al. [47] extend hybrid additive noise models to mixtures, enabling causal direction identification and clustering of mechanism expressions across environments. These developments are directly relevant to this thesis, which seeks to extend such mechanism clustering to settings with unknown numbers of regimes and without temporal ordering.

2.2.3 Causal Discovery for Time Series (Stationary and Non-stationary)

Assaad et al. [36] categorize time-series causal discovery into Granger-based, constraint-based, noise-based, score-based, logic-based, topology-based, and difference-based families, and empirically compare them on synthetic and real data. Their conclu-

sion is that no single family dominates universally, instead each relies on assumptions such as linearity, causal sufficiency, stationarity, or acyclicity that may or may not hold in a given dataset. This motivates the need for methods that can flexibly handle heterogeneous data without strong distributional commitments. Glymour et al. [26] further emphasise that multivariate time series introduce severe compounding challenges absent in their i.i.d. counterparts, most notably nonlinearity, nonstationarity, subsampling, and unmeasured confounding.

Addressing the specific challenge of nonstationarity, Sadeghi et al. [49] recently introduced CD-NOTS (Constraint-based Causal Discovery from Nonstationary Time Series), building directly upon the foundational CD-NOD framework [29]. CD-NOTS is a nonparametric algorithm designed to explicitly model both contemporaneous and time-lagged effects while enforcing causal graph consistency across time. A key flaw in CD-NOD identified by the authors is that its treatment of lagged relationships can yield erroneous graphs when time lags are included in the conditioning set. CD-NOTS corrects this by including lagged variables in all relevant independence tests. To ensure adaptability, the framework supports four conditional independence tests (KCIT, RCoT, CMiknn, and ParCorr) with systematic guidance on test selection across different sample-size and dimensionality regimes. When evaluated on approximately two decades of real-world financial data, including Fama-French factors and Apple stock returns, macroeconomic indices across seven countries, and financial company ratios, CD-NOTS consistently uncovers statistically significant, nonstationary and nonlinear causal relationships that traditional stationary or linear methods fail to detect. These findings highlight the dangers of model misspecification and provide a powerful practical motivation for nonstationary-aware causal discovery.

2.2.4 Deep-learning and Granger-based Methods

Recent years have witnessed a surge in integrating deep learning architectures to extend traditional Granger causality to high-dimensional and non-linear domains. For instance, Nauta et al. [28] introduced the Temporal Causal Discovery Framework (TCDF), which leverages attention-based temporal Convolutional Neural Network (CNN) alongside a permutation-based validation step to discover both temporal causal graphs and explicit time lags from observational time series. Absar et al. [38] (NTiCD) use LSTMs and graph convolutions to learn summary graphs consistent with Granger causality. To target high-dimensional and irregularly sampled streams, Cheng et al. [40] introduces CUTS+, which couples Granger-based neural components with a coarse-to-fine discovery mechanism powered by message-passing Graph Neural Networks (GNNs). More recently, Zhu et al. [50] shifted the focus toward non-stationary multivariate environments by directly mapping Transformer attention matrices to dynamically evolving Granger causal graphs.

These methods operate in a fundamentally sequential manner, exploiting autoregressive temporal dependencies. This structure does not have an equivalent in unordered independent and identically distributed (i.i.d.) observations.

2.2.5 Temporal Causal Discovery with Unknown Regimes: CASTOR

The method most closely related to this thesis is CASTOR (Causal Temporal Regime Structure Learning), introduced by Rahmani and Frossard [48], published at AIS-TATS 2025. CASTOR is designed to learn causal relationships from multivariate time series composed of multiple, unknown sequential regimes, concurrently identifying three critical unknowns: the total number of regimes, their sequential boundaries (start and end time points), and the corresponding causal DAG underlying each regime.

Methodologically, CASTOR employs an Expectation-Maximisation (EM) framework. In the E-step, the algorithm updates posterior probabilities of regime membership based on current causal graphs. In the M-step, it learns each per-regime DAG by optimising a BIC-penalised score subject to the NOTEARS acyclicity constraint [24]. A key design feature is that CASTOR requires no prior specification of the number of regimes; it initialises by over-partitioning the time series into equal-width windows and iteratively merges adjacent segments that exhibit similar causal structures until convergence. The authors establish theoretical identifiability for both the ground-truth regimes and their respective DAGs up to a permutation, assuming Gaussian noise with equal variance.

Experimentally, CASTOR is evaluated on synthetic datasets up to 40 nodes as well as on two real-world benchmarks: IT infrastructure monitoring data and biosphere-atmosphere gas exchange data. Across these tasks, it consistently outperforms CD-NOD [29], RPCMCI, and DYNOTEARS [33] in regime detection accuracy (measured by Adjusted Rand Index) and causal graph recovery (measured by SHD and F1).

CASTOR is the closest existing method to the framework proposed here, and the comparison is instructive in three respects.

First, both methods target the same problem: joint discovery of an unknown number of sequential regimes and their regime-specific causal graphs, from unlabelled multivariate time series, without environment labels or a surrogate nonstationarity index. This shared target distinguishes both from ICP, IRM, and JCI (which require labelled environments) and from CD-NOD and CD-NOTS (which require an auxiliary index encoding the changing conditions).

Second, the methods differ fundamentally in how a candidate regime boundary is evaluated. CASTOR optimises a BIC-penalised likelihood within an expectation-maximisation loop, initialising by over-partitioning into equal-width windows and merging adjacent segments with similar causal structure. A boundary is therefore accepted or rejected on the basis of a global score, and the procedure offers no explicit hypothesis test for whether the mechanisms on either side of a proposed boundary differ significantly. LIRA instead evaluates each candidate boundary with a formal statistical test – the Chow F -test on the coefficient vectors of the structural model – before any complexity criterion is applied. This yields an interpretable p -value per boundary and addresses a gap the literature has identified but not closed: the absence of formal, testable criteria for determining whether two identified regimes are structurally different rather than artefacts of finite-sample variation.

Third, LIRA's structural model is not generic. It is derived from the power-law physics of the Fundamental Plane of Black Hole Activity (Section 2.1.2), which fixes

the functional form, the variable set, and consequently the degrees of freedom of the test. CASTOR is domain-agnostic by design. The trade-off is explicit: LIRA sacrifices generality for a physically motivated model in which every parameter has an interpretation, and in which the identifiability of the recovered structure can be established from the model form itself (Section 4.6).

Two limitations of the present approach relative to CASTOR should be stated. LIRA's binary segmentation is greedy, and can in principle miss configurations in which two boundaries are only jointly detectable, whereas CASTOR's EM procedure optimises regime assignment globally. LIRA also assumes piecewise-constant structure with hard boundaries; DyCAST's Neural ODE formulation models causal structure as evolving continuously, which may better approximate accretion dynamics that are in reality gradual. Both are identified as directions for future work.

2.2.6 Continuously Evolving Causal Structure: DyCAST

A complementary frontier result is DyCAST [44], published at ICLR 2025. While CASTOR models causal structure as piecewise constant across discrete sequential regimes, DyCAST models it as evolving continuously over time. it achieve this by using Neural ODEs constrained to remain on the DAG manifold at all times, successfully capturing smooth causal transitions that discrete regime boundaries would poorly approximate.

When evaluated across synthetic dynamic datasets, fMRI data, the CausalTime benchmark, and human motion capture data, DyCAST consistently outperforms all static baselines and scales substantially beyond methods like DYNOTEARS [33], NTS-NOTEARS, and TECDI, all of which fail entirely when processing 50 or more variables. This scalability finding is highly significant because it confirms that even purpose-built temporal methods struggle at moderate dimensionality. This limitation directly motivates our own scalability analysis in i.i.d. causal clustering (see Section 2.4.4).

Like CASTOR, DyCAST relies fundamentally on temporal ordering. Its Neural ODE framework parameterises causal evolution strictly as a function of time, leaving it with no functional analogue for unordered observations. Consequently, while both CASTOR and DyCAST provide strong temporal motivation and methodological inspiration, neither offers a direct solution for the i.i.d. setting addressed in this thesis.

2.2.7 Challenges and Methodological Themes

A review of recent literature demonstrates that the performance of causal discovery methodologies remains closely tied to their underlying structural assumptions. As-saad et al. [36] stress that performance depends on how closely the target dataset adheres to assumptions such as linearity, causal sufficiency, and time invariance. A recent innovation, such as the internal incoherency scores introduced by Faltenbacher et al. [45] provides a promising path forward by dynamically flagging model misspecifications and finite-sample errors. When these assumptions are violated, performance degrades significantly. Simultaneously, managing high dimensionality and mixed data types remains as a central algorithmic hurdle. Researchers have approached this challenge from multiple angles, including hybrid Bayesian network

learning [22], robust constraint-based conditional independence (CI) tests tailored for mixed data domains [23], and coarse to fine discovery mechanisms (CUTS+) embedded within graph neural networks [40].

Parallel to these algorithmic optimizations, modern causal discovery increasingly prioritizes rigorous uncertainty quantification over rigid point-estimate graph reconstructions. Rather than outputting a single point estimate graph, variational causal networks now approximate full posterior distributions over possible DAG structures [34]. Similarly, joint DAG-effect Bayesian allow structural uncertainty to propagate directly into downstream intervention effect estimates [35], ensuring that subsequent causal inferences reflect the true ambiguity of the learned graph. Finally, there is a clear shift toward validating causal discovery. Zhang et al. [41] shows that learned Granger graphs can directly improve forecasting performance, linking causal discovery to downstream tasks.

2.2.8 Conclusion of the Survey

Over two decades, causal discovery has progressed from classical DAG learning to a rich toolbox handling mixed data, heterogeneity, uncertainty, and temporal dynamics. For time series, work spans Granger-based statistics, deep neural methods, constraint/score-based extensions, and most recently published methods such as Rahmani and Frossard [48] and Sadeghi et al. [49] that jointly discover sequential regimes and their causal graphs while handling nonstationarity. No single method is universally best; performance depends critically on how well model assumptions This thesis targets a setting that remains underexplored even within this recent literature: the unsupervised clustering of causal mechanisms from independent and identically distributed (i.i.d.) observational data. We address this problem without relying on temporal structures, environment labels, or prior knowledge regarding the true number of regimes. While Rahmani and Frossard [48] provides a temporal analogue and key methodological inspiration for this objective, its reliance on sequential ordering means it cannot offer a direct solution. Developing an analogous joint framework for unordered i.i.d. observations constitutes the central contribution of this work.

Table 2.1: Summary of major related works in causal discovery, regime detection, and causal clustering.

Study	Task	Method	Setting	Main Finding / Gap
Spirtes et al. [6]	DAG learning from observational data	PC and FCI constraint-based algorithms	i.i.d., single environment	Foundational framework for causal discovery; assumes a single fixed causal model throughout.
Zheng et al. [24]	Continuous DAG structure learning	NOTEARS: smooth algebraic acyclicity constraint	i.i.d., single environment	Enables gradient-based DAG learning at scale; acyclicity constraint underpins DYNOTEARS and CASTOR.
Continued on next page				

Table 2.1 – continued from previous page

Study	Task	Method	Setting	Main Finding / Gap
Pamfil et al. [33]	Time-series causal graph recovery	DYNOTEARS: NOTEARS extended to contemporaneous and lagged edges	Stationary time series	Efficiently recovers window causal graphs; assumes stationarity holds throughout the entire series.
Huang et al. [29]	Causal discovery under heterogeneity	CD-NOD: nonparametric method exploiting distribution shifts	Multiple domains or time-indexed data	Heterogeneity aids causal identification; requires auxiliary domain index or time stamp.
Peters et al. [20]	Causal parent identification	ICP: invariant prediction across environments	Multiple labelled environments	Exploits causal invariance effectively; requires environment labels as a strict prerequisite.
Arjovsky et al. [25]	Out-of-distribution generalisation	IRM: invariant risk minimisation	Multiple labelled environments	Encourages causally stable features; guarantees weaken when environment count is small.
Perry et al. [37]	Multi-environment causal graph recovery	Sparse Mechanism Shift hypothesis with MSS score	Multiple labelled environments	Full graph identifiable under sparse shifts; environment membership must still be known.
Huang et al. [27]	Causal clustering with shared and specific relations	SSCM: mixture of causal mechanisms	i.i.d., multiple regimes	Distinguishes shared and regime-specific causal relations; requires pre-specified regime count.
Chen et al. [39]	Causal structure learning from unknown environments	CCSL: Bayesian nonparametric Causal CRP with variational inference	i.i.d., unknown environments	Does not require regime count in advance; identifiable under linear non-Gaussian assumption. Primary baseline for this thesis.
Varambally et al. [42]	Mixture of causal models from time series	MCD: variational inference over mixtures of SCMs	Temporal, autoregressive structure	Jointly infers causal models and mixing probabilities; relies on temporal ordering.
Continued on next page				

Table 2.1 – continued from previous page

Study	Task	Method	Setting	Main Finding / Gap
Rahmani and Frossard [48]	Temporal causal regime discovery	CASTOR: EM over piecewise-constant regimes with NOTEARS acyclicity	Sequential time series	Jointly learns regime boundaries and per-regime DAGs without prior regime count; relies on temporal contiguity unavailable in i.i.d. settings.
Cheng et al. [44]	Continuously evolving causal structure	DyCAST: Neural ODE constrained to DAG manifold	Sequential time series	Captures smooth causal transitions and scales beyond 50 nodes; requires temporal ordering.
Li et al. [46]	Unsupervised causal clustering from mixed-type data	HCL: bi-directional iterative causal clustering (preprint)	i.i.d., no environment labels	No temporal ordering or regime count required; scalability on large graphs not yet assessed.
Sadeghi et al. [49]	Causal discovery from nonstationary time series	CD-NOTS: nonparametric extension of CD-NOD correcting lagged conditioning	Non-stationary time series	Uncovers nonlinear nonstationary causal relations missed by stationarity-assuming methods.
Mooij et al. [31]	Joint causal inference from multiple contexts	JCI: context encoded as additional graph nodes	Multiple pre-specified contexts	Theoretically general and subsumes many special cases; contexts must be labelled in advance.
Faltenbacher et al. [45]	Assumption violation detection	Internal incoherency scores for constraint-based algorithms	i.i.d., single environment	Flags assumption violations without ground truth; not yet adapted to causal clustering settings.

2.3 Foundations of Causal Discovery

Causal discovery refers to inferring causal relationships among variables directly from observational data, without controlled experiments. The field has its roots in the constraint-based framework of Spirtes et al. [6], which represents causal structure as a directed acyclic graph (DAG) and recovers it by evaluating statistical conditional independence relationships. Within the paradigm, the PC algorithm begins with a fully connected graph and removes edges where conditional independence is detected, producing a Markov equivalence class of DAGs. The FCI algorithm

extends PC to allow for latent confounders, producing a partial ancestral graph (PAG).

Score-based methods search over the space of DAGs to find the structure that best fits the data according to some criterion. The Greedy Equivalence Search (GES) performs a two-phase greedy search over equivalence classes. Zheng et al. [24] (NOTEARS) reformulated structure learning as a continuous optimisation problem by expressing acyclicity as a smooth algebraic condition, enabling gradient-based DAG learning at scale. This formulation underpins many derivative methods, including DYNOTEARS [33], which extends NOTEARS to handle both contemporaneous (intra-slice) and time-lagged (inter-slice) dependencies within a structural equation model (SEM) framework.

Functional causal models (FCMs) make explicit assumptions about functional form to achieve full identifiability of causal direction, a resolution that constraint-based methods cannot achieve from a single observational distribution. The most prominent example, Shimizu et al. [14], shows that if the data-generating process is linear and noise terms are non-Gaussian, the full causal order is uniquely identifiable via Independent Component Analysis. Identifiability here means full causal ordering recovery under the linear non-Gaussian assumption. This result extends to the time series domain through VAR-LiNGAM [17], which combines the linear non-Gaussian assumption with vector autoregressive modelling to handle both instantaneous and lagged causal effects.

Despite their contributions, all of these foundational methods share a critical assumption: that the entire dataset originates from a single, fixed causal model. Real-world data frequently violates this homogeneity condition. Empirical data often arise from multiple distinct latent processes, such as varying experimental conditions, heterogeneous population subgroups, shifting temporal intervals, or differing biological states, where each group is governed by its own unique causal structure. When such mixed-mechanism data are processed by a traditional single-environment algorithm, the optimization routinely collapses into a spurious, averaged graph that fails to accurately represent any of the true underlying systems. This critical limitation highlights the need for heterogeneous structure learning and provides the primary motivation for the research directions evaluated in the following sections.

2.3.1 Causal Discovery from Nonstationary and Heterogeneous Data

A substantial body of research extends causal discovery to settings where the data-generating process changes across domains, time periods, or experimental conditions, exploiting the insights that such nonstationarity and heterogeneity carry structural information can improve causal identification.

The foundational contribution is Invariant Causal Prediction (ICP) [20], which identifies the direct causal parents of a target variable as the minimal predictor set whose conditional relationship with the target remains invariant across known environments. This elegantly exploits the stability of causal mechanisms, but requires environment labels to be provided in advance.

Huang et al. [29] developed CD-NOD, which exploits distribution shifts across domains or over time to improve causal skeleton recovery and edge orientation. A key insight is that data heterogeneity benefits causal identification: shifting mecha-

nisms leave detectable fingerprints that break Markov equivalence class ambiguity. CD-NOD requires an auxiliary variable (domain index or time stamp) to model nonstationarity, but not hard environment labels.

Sadeghi et al. [49] introduced CD-NOTS as a direct extension of CD-NOD for time series, correcting its treatment of lagged relationships and providing principled guidance for conditional independence test selection (see Section 2.2 for details). Perry et al. [37] advanced the theory through the Sparse Mechanism Shift hypothesis: if only a small number of causal mechanisms change between environments, the full causal graph becomes identifiable with high probability. But it again requires known environment membership.

In the temporal domain, Cheng et al. [44] (ICLR 2025) models the continuous temporal evolution of the causal DAG using Neural ODEs (see Section 2.2 for the detailed comparison with CASTOR). Its strong scalability and performance results deliver a clear lesson: real-world causal structures are not static, and methods that model this dynamic nature recover substantially more accurate and interpretable causal graphs.

Arjovsky et al. [25] introduced Invariant Risk Minimisation (IRM) as a learning-theoretic complement to ICP, framing multi-environment causal learning as an out-of-distribution generalisation problem. IRM encourages reliance on causally stable features, but requires known environment partitions and its guarantees weaken when the number of environments is small relative to the number of features.

Taken together, these works establish a clear principle: heterogeneity and nonstationarity are assets, not obstacles, for causal learning. However, they all require either environment labels, an auxiliary nonstationarity index (CD-NOD [29], CD-NOTS [49]), or temporal ordering (CASTOR [48], DyCAST [44]). Recovering regime structure from purely i.i.d. data without any such external information is the central gap this thesis targets.

2.3.2 Mixture Models and Causal Clustering

A smaller but rapidly growing body of work addresses simultaneous clustering and causal structure learning, targeting scenarios where the environment partition must itself be inferred. These works recognise that clustering and causal discovery must be solved jointly: sequential approaches suffer from error propagation, since incorrect cluster assignments corrupt causal structure learning and vice versa.

An early and influential contribution is the Specific and Shared Causal Model (SSCM) framework of Huang et al. [27], which models data as arising from a mixture of causal mechanisms and clusters observations by the mechanism that generated them, distinguishing regime-specific from shared causal relations. However, SSCM requires the number of regimes to be specified in advance.

Chen et al. [39] proposed CCSL (Causal Clustering Structure Learning), published in *IEEE Transactions on Neural Networks and Learning Systems*, and the most directly relevant prior work to this thesis. CCSL addresses learning causal structure from non-i.i.d. data collected from multiple unknown environments, where neither the number of environments nor their assignment is known. The key innovation is a Causality-related Chinese Restaurant Process (Causal CRP), a Bayesian non-parametric clustering procedure assigning observations to clusters based on causal structure similarity rather than feature-space distance. A variational inference ap-

proach simultaneously estimates the structural equation model parameters for each cluster, and both components are updated within a single joint objective. CCSL does not require the number of groups to be specified: the Bayesian nonparametric prior grows the number of clusters as needed. Theoretical identifiability of both the causal model and the clustering assignment is established under the linear non-Gaussian assumption, matching the condition required by traditional LiNGAM architectures [14]. Experiments on fMRI and Sachs protein signalling data validate the method on real scientific problems. CCSL serves as the primary baseline and conceptual reference for the approach developed in this thesis.

i. Temporal Extensions of Causal Clustering

Varambally et al. [42] proposed MCD (Mixture of Causal Discovery), a variational inference framework for discovering mixtures of structural causal models from time series data. MCD infers both the underlying causal models and the per-sample mixing probabilities end-to-end, with MCD-Linear and MCD-Nonlinear variants. Its reliance on autoregressive temporal structure makes it inapplicable to i.i.d. cross-sectional data.

As detailed in Section 2.2, Rahmani and Frossard [48] (AISTATS 2025) represents the current state of the art in temporal causal regime discovery. Its EM-based joint discovery of unknown sequential regimes and their causal DAGs, combined with identifiability guarantees under Gaussian equal-variance noise, provides the strongest existing evidence that causal regime discovery is both principled and practical. The crucial distinction for this thesis is that CASTOR relies strictly on temporal contiguity to propagate its soft regime assignments. When temporal ordering is absent, this assignment mechanism becomes unavailable, necessitating an entirely different approach for static, independent and identically distributed data.

ii. i.i.d. Extensions and Recent Advances

Seeking to address unordered data directly, Li et al. [46] proposed HCL (Heterogeneous Causal structure Learning), an unsupervised framework that jointly infers latent clusters and their causal structures from mixed-type observational data, requiring no temporal ordering, environment labels, interventions, or prior knowledge of the number of regimes. HCL relaxes homogeneity and causal sufficiency assumptions via an equivalent representation encoding both structural heterogeneity and confounding, and uses a bi-directional iterative strategy to alternate between causal clustering and structure learning. Identifiability is established under conditions weaker than the linear non-Gaussian requirement of CCSL, though the precise conditions are not fully spelled out in the current preprint. Evaluated on synthetic datasets and single-cell perturbation data, HCL demonstrates the practical utility of mechanism-level clustering. Questions remain regarding scalability, however: experiments in the preprint use relatively small graphs, and performance at the node counts (40+) evaluated in CASTOR [48] and DyCAST [44] has not been assessed. As a preprint, HCL's identifiability claims and empirical results should be treated with appropriate caution pending peer review.

Liu et al. [47] extend hybrid additive noise models to mixtures, proposing the Mixture Conditional Variational Causal Inference (MCVCI) model and the MCVCC

clustering method, showing that even simple bivariate settings benefit from mechanism-aware clustering.

iii. Classical Clustering Methods

Classical methods including k-means, Gaussian Mixture Models (GMMs), DBSCAN, and OPTICS, define clusters in terms of distributional similarity in observation space and are systematically outperformed by causally-aware methods when the true cluster structure reflects differences in causal mechanism rather than feature-level proximity, as demonstrated empirically in CCSL [39]. Integrating causal structure into the clustering objective is therefore not a refinement but a fundamentally different approach to the problem.

2.3.3 Joint Causal Inference from Multiple Contexts

Mooij et al. [31] developed Joint Causal Inference (JCI), a unifying graphical framework for causal discovery from multiple observational and interventional contexts. JCI encodes context as additional nodes in the causal graph, allowing both causal structure and intervention type to be inferred jointly. The approach is theoretically general and subsumes many multi-environment special cases, but presupposes that the contexts themselves are known and pre-specified, placing it in the same category as other labelled-environment methods.

The broader causal representation learning literature has explored disentangling latent causal variables from high-dimensional observations via variational autoencoders or other deep generative models. These methods share the goal of uncovering hidden causal structure, but typically focus on identifying latent variables rather than partitioning observations by causal regime, and are evaluated predominantly in image or video domains rather than tabular scientific data.

2.4 Summary of Key Findings

The literature reviewed reveals a clear progressive trajectory. The discipline has advanced from single-environment causal discovery, through multi-environment methods that exploit known partitions or auxiliary nonstationarity indices, to joint inference of causal structure and data clustering without prior environment information. This section consolidates key methodological trends, consistent findings, contradictions and limitations, and articulates the specific research gaps addressed by this thesis alongside an explicit statement of scope.

2.4.1 Methodological Trends

Within foundational causal discovery, constraint-based methods (PC, FCI) and score-based methods (GES, NOTEARS [24], DYNOTEARS [33]) represent the two primary paradigms, with FCMs (LiNGAM [14], VAR-LiNGAM [17], ANMs) occupying an assumption-heavy but important niche. As surveyed by Assaad et al. [36], these families differ in their core assumptions: Granger causality relies on predictive precedence; constraint-based methods exploit conditional independence; noise-based methods leverage distributional assumptions (e.g. non-Gaussianity); and score-based

methods optimise a global criterion such as BIC. Neural extensions (TCDF [28], NTiCD [38], CUTS+ [40], [50]) demonstrate that neural architectures can learn causal structure from high-dimensional or non-stationary time series. As noted in Section 2.2.4, these methods are inherently sequential and inapplicable to the i.i.d. setting of this thesis.

In multi-environment and nonstationary research, the use of invariance or distributional shift as a guiding principle is the defining methodological commitment. This commitment manifests across diverse implementations, including statistical invariance tests (ICP [20]), distributional shift detection (CD-NOD [29], CD-NOTS [49]), learning objectives (IRM [25]), or graphical independence constraints (JCI [31]). The Sparse Mechanism Shift hypothesis [37] formalises the theoretical bedrock of these approaches, proving that if only a few mechanisms change across environments, the full causal graph becomes identifiable with high probability. The CD-NOTS study [49] further demonstrates the practical cost of assumption misspecification.

In the temporal domain, CASTOR [48] and DyCAST [44] demonstrate that explicitly modelling temporal causal dynamics, configured either as piecewise constant regimes or as continuously evolving structures respectively, yields substantially more accurate results. CASTOR's EM-based joint learning establishes that alternating between soft regime assignment and causal graph estimation is theoretically principled and empirically superior.

In the causal clustering literature, Bayesian nonparametric methods, especially Chinese Restaurant Process-based approaches are the dominant strategy for jointly inferring cluster assignments and causal structures [27, 39]. CCSL [39] demonstrates joint inference is practically superior to sequential pipelines. HCL [46] extends these ideas to mixed-type data without temporal ordering. Temporal extensions such as MCD [42] and CASTOR [48] adapt these principles via EM or variational inference. At the bivariate level, Liu et al. [47] provide a complementary perspective on mechanism-aware clustering.

2.4.2 Consistent Findings

Several consistent findings emerge across the reviewed works.

1. Causal structure varies substantially across regimes; methods ignoring this heterogeneity produce spurious averaged graphs that accurately represent none of the true underlying mechanisms [29, 39, 42, 46, 48].
2. Joint inference of cluster assignments and causal structures consistently outperforms sequential pipelines [27, 39, 46, 48].
3. Causal structure similarity is a more stable clustering signal than feature-space distance or correlation, as demonstrated by the CCSL experiments [39], where causally-aware clustering substantially outperforms k-means, DBSCAN, and OPTICS across all tested conditions.
4. Real-world data consistently violates the stationarity assumption, as shown by CD-NOTS [49] across financial case studies and by CASTOR [48] in biosphere-atmosphere and IT monitoring data.
5. No single causal discovery method is universally best; performance depends critically on whether the method's assumptions match the data-generating process [36].

2.4.3 Contradictions and Limitations

Invariance-based multi-environment methods (ICP [20], IRM [25]) require known environment partitions. When environment labels are missing or incorrect, their theoretical guarantees collapse, and no consensus exists on how to robustly recover environment structure in a purely unsupervised manner.

In the temporal domain, a tension exists between methods designed for stationary data and those for nonstationary data. CD-NOTS [49] shows that applying stationarity-assuming algorithms to real-world financial data produces spurious causal graphs. Yet many causal clustering methods, including CCSL [39], assume stationarity within each cluster. While CASTOR [48] handles sequential nonstationarity, it assumes piecewise-constant regimes and does not address i.i.d. clustering without temporal order. Its identifiability guarantees further require Gaussian equal-variance noise, which may be overly restrictive.

In the causal clustering literature, the core tension is between model expressivity and identifiability. CCSL [39] provides rigorous identifiability under the linear non-Gaussian assumption (full causal ordering recovery, as in LiNGAM [14]), but this may be too restrictive for nonlinear data. More expressive frameworks such as HCL [46] and MCD [42] relax these assumptions but sacrifice formal identifiability guarantees. For instance, the HCL claims identifiability under “mild conditions” not fully specified in the current preprint. CASTOR [48] provides identifiability for temporal data under Gaussian equal-variance assumptions. Liu et al. [47] offer a potential bivariate middle ground, but the appropriate trade-off between expressivity and identifiability for unsupervised causal clustering remains open.

Evaluation methodology is also inconsistent across works: some papers measure clustering quality independently of causal graph recovery; others report only graph accuracy. The absence of a unified benchmark makes cross-method comparison difficult. Internal incoherency scores [45] offer a promising direction for assumption-violation detection, but have not yet been adapted to the causal clustering setting.

2.4.4 Research Gaps

The reviewed literature reveals several notable gaps. For each gap, the scope with respect to this thesis is stated explicitly.

- Absence of a formal invariance test in temporal regime discovery.** Methods requiring labelled environments (ICP, IRM, JCI) or an auxiliary nonstationarity index (CD-NOD, CD-NOTS) are inapplicable where the latent state is itself unknown. Among methods that discover regimes jointly with causal structure from unlabelled series, CASTOR and DyCAST represent the state of the art, but neither subjects a proposed regime boundary to an explicit statistical hypothesis test; boundaries are accepted on the basis of a global optimisation score. The consequence is that a discovered boundary carries no associated significance statement. This thesis addresses that gap directly by constructing a boundary criterion from the Chow test for equality of regression coefficients, extended to accommodate temporal autocorrelation, and by pairing it with an independent complexity criterion so that statistical significance and model parsimony are evaluated as separate conditions.

- **Lack of formal criteria for structural difference.** Existing works, including CCSL [39] and CASTOR [48], do not provide formal, testable criteria for determining whether two identified regimes are structurally different rather than statistical artefacts of finite-sample variation.
- **Limited scalability evaluation.** Most causal clustering methods have been evaluated on small synthetic datasets. DyCAST [44] demonstrates that even purpose-built temporal methods can struggle at scale; CASTOR [48] evaluates up to 40 nodes, but comparable analysis for i.i.d. causal clustering methods is largely absent.
- **No standard benchmark for causal clustering.** Unlike single-environment causal discovery which benefits from established benchmarks such as SACHS and DREAM challenges, causal clustering lacks agreed-upon evaluation datasets, making progress difficult to measure.
- **Underdeveloped theory for observational identifiability.** Identifiability guarantees for the unsupervised mixture-of-SCMs problem from purely observational i.i.d. data are largely absent beyond the linear non-Gaussian case of CCSL [39]. CASTOR provides identifiability only for temporal data under Gaussian equal-variance assumptions.
- **Handling of nonstationarity within regimes.** Existing methods, including CCSL [39] and CASTOR [48], assume stationarity within each regime. A framework handling within-regime nonstationarity alongside regime discovery would be a significant advance.
- **Assumption violation detection without ground truth.** Faltenbacher et al. [45] introduce incoherency scores for constraint-based algorithms, but no analogous tools exist for causal clustering methods.

These gaps collectively define a significant and tractable research opportunity. The primary contribution of this thesis, which a principled framework for unsupervised clustering of causal mechanisms from i.i.d. observational data, covering regime discovery, structural comparison, and per-regime causal graph reconstruction, directly targets the first, second, third, and fifth gaps above. The remaining gaps (standardised benchmarks, within-regime nonstationarity, assumption violation detection) are acknowledged as important directions for future work.

Chapter 3

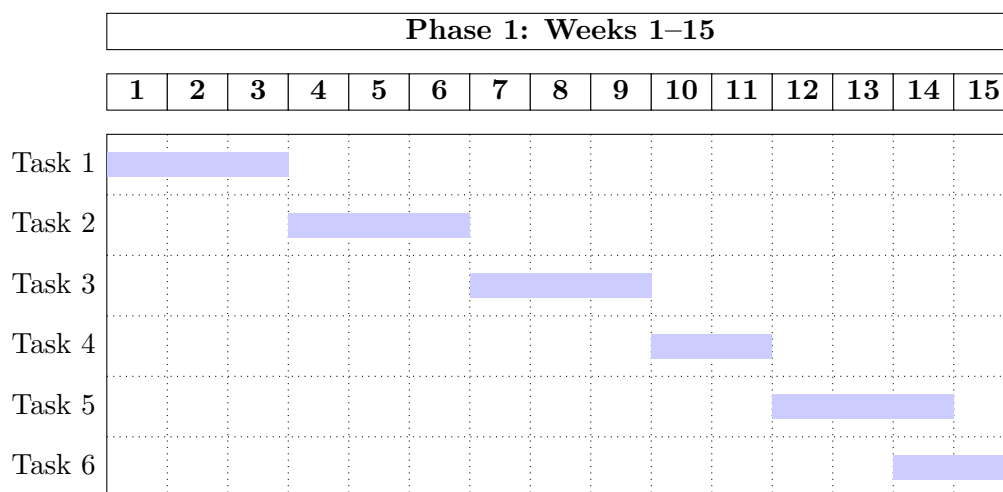
Requirements, Impacts and Constraints

3.1 Project Management Plan

The research project is split into three phases over about 46 weeks.

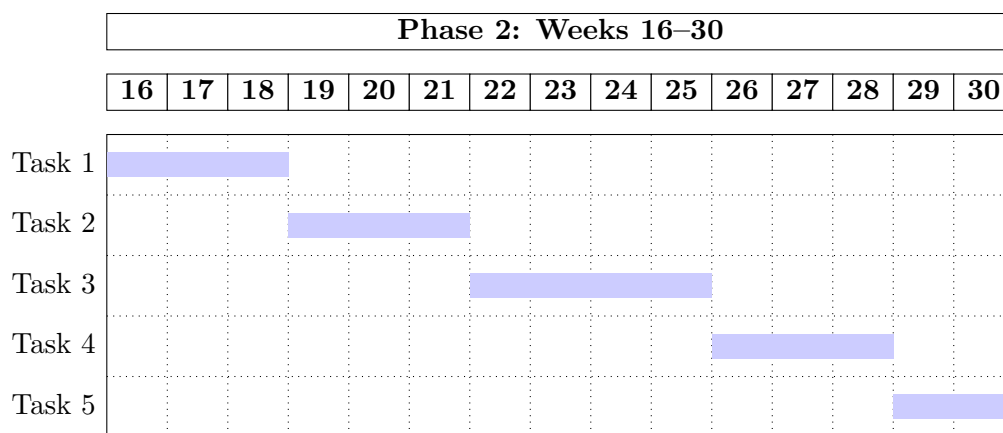
Phase 1 (Weeks 1–15): Background and Methodology Design

1. Review time-series causal discovery methods.
2. Review unsupervised clustering for regime identification.
3. Review accretion disk physics and known accretion states (low/hard, high/soft, quiescent).
4. Identify gaps in existing causal discovery for astrophysical time-series.
5. Design the regime separation and causal graph recovery pipeline.
6. Obtain or simulate accretion disk time-series datasets (e.g., from GRMHD simulations or public archives like NICER/RXTE).



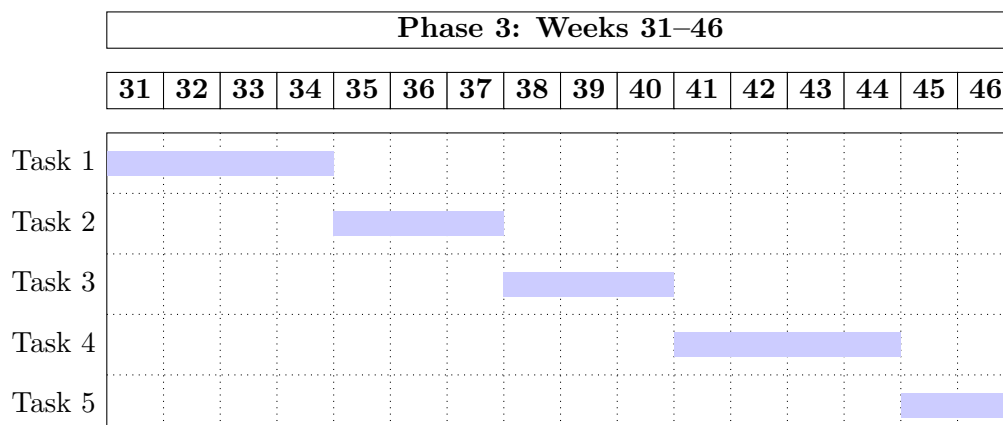
Phase 2 (Weeks 16–30): Algorithm Development and Testing on Simulated Data

1. Generate simulated data with known ground-truth causal graphs and known regime labels.
2. Implement regime clustering based on causal mechanism shifts.
3. Implement time-series causal discovery for each estimated regime.
4. Validate the pipeline on simulated data by comparing recovered graphs to the ground truth.
5. Perform sensitivity analysis (noise levels, sample size, irregular sampling).



Phase 3 (Weeks 31–46): Application to Real Observations and Reporting

1. Apply the pipeline to real accretion disk light curves (e.g., from GRS 1915+105 or Cygnus X-1).
2. Interpret the discovered regimes in terms of known accretion states.
3. Compare causal graphs across regimes and discuss physical interpretation.
4. Draft the final report, including methodology, results, and limitations.
5. Prepare visualizations (causal graphs, regime assignments, time-series plots).



3.2 Risk Management

The biggest risk in this thesis is that real astrophysical time-series data from black hole accretion disks, like X-ray light curves, often have irregular patterns, noise, and limited quantity. This makes it hard to reliably discover causal graphs and separate unknown regimes.

Unlike controlled lab experiments, it is impossible to label which accretion state produced each data point, and the actual number of distinct causal mechanisms in the data is unknown. Also, hidden or unobserved variables, such as changes in the magnetic field, may act as confounders, creating false causal connections between the measured features. Many standard causal discovery algorithms assume there are no cycles or feedback loops, but accretion disks show strong feedback between variables like temperature, density, and accretion rate.

To manage these risks, the research will rely primarily on simulated data where the true causal graphs are known. It will also use algorithms designed for time-series data that handle time delays and hidden variables. The results will be compared across different methods to ensure stability against random noise and small sample sizes.

Chapter 4

Proposed Methodology

4.1 Overview

The framework operates as a four-stage pipeline. **Stage 0** transforms observables into log space, a step derived from accretion physics rather than adopted as a pre-processing convenience (Section 4.2). **Stage 1** discovers latent regime boundaries by recursive binary partitioning, using a lagged Chow F -test as a detection criterion and the Bayesian Information Criterion as an independent stopping rule (Section 4.3–4.5). **Stage 2** recovers a regime-specific dynamic causal graph within each discovered partition using DYNOTEARS (Section 4.7), and **Stage 3** subjects the resulting causal effects to a formal validation battery (Section 4.8).

Figure 4.3 presents the recursive decision pipeline. Figure 4.1 illustrates the physical states the method is designed to distinguish; the corresponding hypothesised causal structures are shown in Figure 4.2, while the realised recursion is presented with the validation results in Chapter 5.

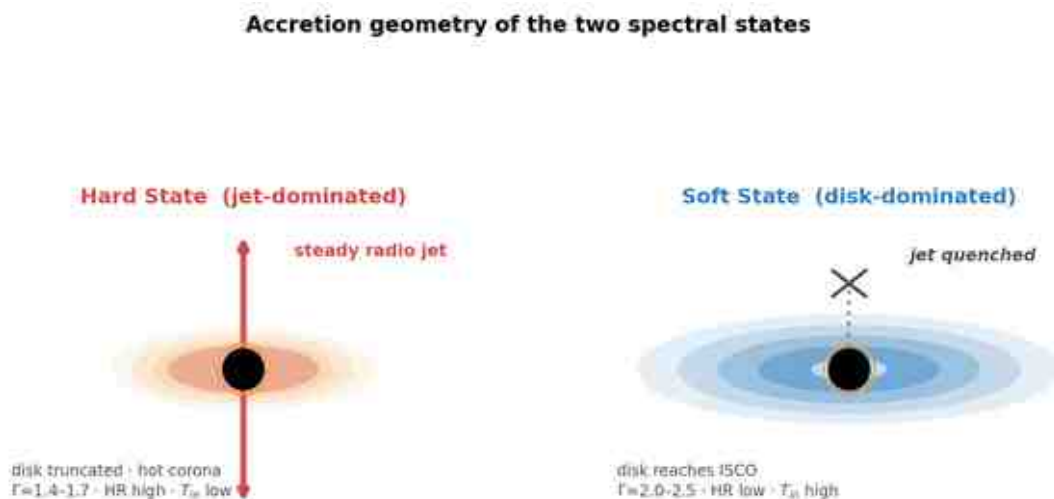


Figure 4.1: Accretion geometry of the two spectral states. In the hard state the disk is truncated and replaced inward by a hot corona, with a steady radio jet; in the soft state the disk extends to the innermost stable circular orbit and the jet is quenched.

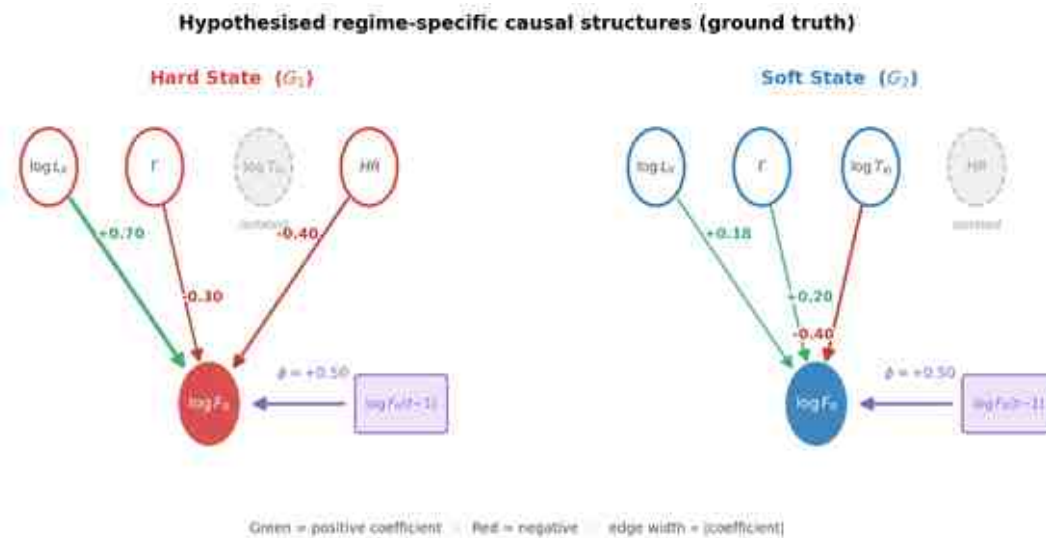


Figure 4.2: Hypothesised causal structures for the hard and soft accretion states.

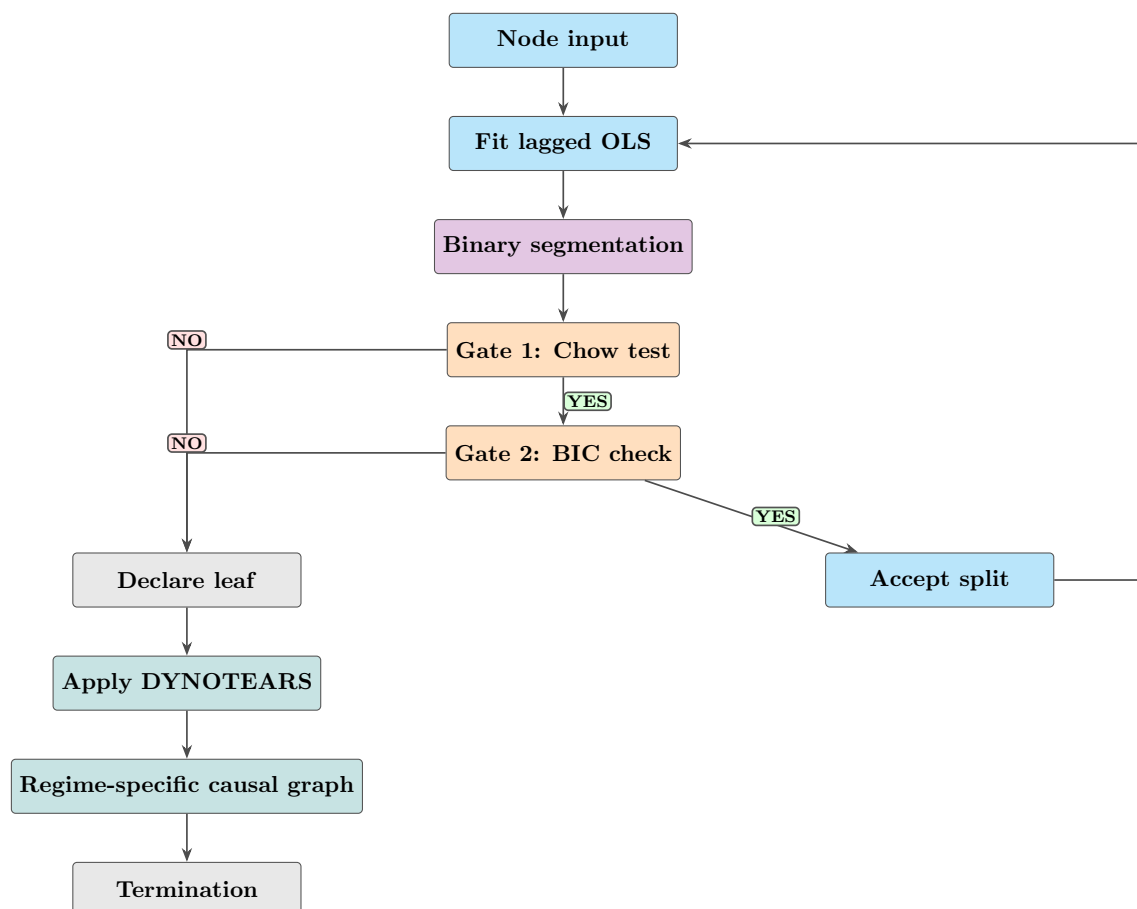


Figure 4.3: Step-by-step recursive framework for temporal regime discovery and dynamic causal graph recovery, combining lagged OLS estimation, Binary Segmentation, Chow and BIC gates, and DYNOTEARS.

4.2 Log-space linearisation

The Fundamental Plane of Black Hole Activity expresses radio luminosity as a power law in X-ray luminosity and black hole mass:

$$L_R \propto L_X^{\xi_X} \cdot M_{BH}^{\xi_M} \quad (4.1)$$

Taking logarithms converts exponents into coefficients and multiplication into addition:

$$\log L_R = \xi_X \log L_X + \xi_M \log M_{BH} + c \quad (4.2)$$

Equation (4.2) is linear. This matters because every subsequent component of the framework rests on linear model theory: ordinary least squares estimation assumes a linear relationship; the Chow test compares two OLS coefficient vectors; the F -distribution derivation in Section 4.4 requires the linear form; the BIC in Section 4.5 is computed from the residual sum of squares of a linear fit; and DYNOTEARS recovers a linear structural equation model. Had the analysis remained in raw flux space, these relationships would be multiplicative, OLS estimates would be biased, and the distributional results underpinning both gates would be invalid. The justification is therefore a chain:

$$\text{accretion physics} \Rightarrow \text{power laws} \Rightarrow \text{log-linearity} \Rightarrow \text{OLS valid} \Rightarrow \text{Chow and BIC valid} \quad (4.3)$$

The log transform is a physical necessity, not a preprocessing choice. This distinguishes the method from the generic application of linear regression to astronomical data: its linearity assumption is derived from the known scaling physics of the system.

Not all variables are transformed. Table 4.1 records the treatment of each and the reason.

The complete variable transformation specification is given in Table 4.1.

Table 4.1: Variable set and transformation

Symbol	Quantity	Transform	Justification
F_R	radio luminosity	log	luminosity; power-law coupled; spans orders of magnitude
L_X	X-ray luminosity	log	luminosity; power-law coupled; spans orders of magnitude
T_{in}	inner disk temperature	log	multicolor disk models give power-law scaling with luminosity
Γ	photon index	none	already a dimensionless spectral slope — itself an exponent; range ≈ 1.4 – 2.5
HR	hardness ratio	none	already a bounded dimensionless ratio; same argument as Γ

The exemption of Γ and HR follows the same reasoning in both cases: a quantity that is already dimensionless and confined to a narrow range neither requires nor benefits from logarithmic compression, and the power-law argument of (4.1)–(4.2) does not apply to it.

The spectral-state diagnostics used by the model are summarised in Table 4.2.

4.3 The structural model

4.3.1 Absorption of black hole mass

Equation (4.2) contains $\log M_{BH}$. For a single accreting source observed over time, M_{BH} is constant: it cannot vary between time steps. A constant term in a regression equation is absorbed entirely by the intercept,

$$\xi_M \log M_{BH} \longrightarrow \text{folded into } w_0 \quad (4.4)$$

so mass disappears from the model. This is the correct treatment of a within-source constant rather than an omission, but it has a consequence worth stating: LIRA's coefficients are not directly comparable to the cross-source exponents of the Fundamental Plane. LIRA operates *within* a source; the Fundamental Plane operates *across* sources.

4.3.2 Spectral-state variables

Removing mass leaves the model unable to distinguish physical states. What is required are quantities that vary over time and that index the accretion state at each instant. The standard spectral-state diagnostics serve this role, as summarised in Table 4.2.

Table 4.2: Spectral-state diagnostics and their regime dependence

Variable	Hard state	Soft state	Physical quantity indexed
$\log L_X$	lower	higher	accretion rate
Γ	flat, ≈ 1.4 – 1.7	steep, ≈ 2.0 – 2.5	Comptonisation vs. disk emission
$\log T_{in}$	lower (disk truncated)	higher (disk at ISCO)	disk inner radius / geometry
HR	high	low	spectral hardness

These variables are **not substitutes for mass**. They capture something the Fundamental Plane structurally cannot represent: temporal state variation within a single source. This extension is the conceptual contribution of the model and is stated here explicitly rather than assumed. The resulting static specification is

$$\log F_R(t) = w_0 + w_1 \log L_X(t) + w_2 \Gamma(t) + w_3 \log T_{in}(t) + w_4 HR(t) + \varepsilon(t) \quad (4.5)$$

4.3.3 The autoregressive extension

Equation (4.4) assumes $\varepsilon(t)$ is i.i.d. For accretion time series this is false. These systems possess memory through viscous timescales, correlated turbulence, propagating fluctuations, and persistent jet structures; the observed light curves exhibit red noise and temporal correlation. The consequences of ignoring this are specific:

1. OLS coefficient estimates remain unbiased;
2. standard errors are underestimated;

3. the Chow F -statistic breaks. Its denominator assumes i.i.d. residuals. Under autocorrelation the denominator is artificially small, the F -statistic artificially large, and the procedure detects regime boundaries that do not exist.

The third consequence is decisive: unmodelled autocorrelation does not merely reduce precision but produces false positives in the primary output of the method. The remedy is a first-order autoregressive term,

$$\log F_R(t) = w_0 + w_1 \log L_X(t) + w_2 \Gamma(t) + w_3 \log T_{in}(t) + w_4 HR(t) + \phi \log F_R(t-1) + \varepsilon(t) \quad (4.6)$$

in which ϕ absorbs the temporal dependency, rendering $\varepsilon(t)$ approximately i.i.d. and restoring the validity of the test. Section 5.2 verifies empirically that this is achieved.

The coefficient ϕ is itself physically interpretable as a measure of temporal persistence, and its comparison across discovered regimes constitutes a secondary output of the framework (Section 5.4).

Lag order one is adopted as the minimum sufficient choice. Each additional lag increases q , reduces the residual degrees of freedom $n - 2q$, and raises the minimum admissible segment size, thereby coarsening the resolution at which regimes can be detected. Should residual autocorrelation persist at higher orders, Newey–West standard errors are the intended robustness measure.

4.3.4 Choice of estimator

Regularised regression is excluded. Ridge and Lasso add penalties ($\lambda \|\beta\|_2^2$ and $\lambda \|\beta\|_1$ respectively) that shrink coefficients toward zero, biasing the estimates. Since the Chow test consists entirely of comparing the coefficient vectors of two groups, shrinking both corrupts the comparison: the test would partly detect differences induced by the penalty rather than by genuine mechanism change. Stage 1 therefore requires unbiased estimates and must not regularise.

This is deliberately in contrast with Stage 2, where DYNOTEARS does employ an L_1 penalty. The two stages answer different questions. Stage 2 asks *which edges exist*, for which sparsity is essential; Stage 1 asks *whether two coefficient vectors are equal*, for which bias is fatal.

Polynomial terms are excluded. The log-linear form is derived from power-law physics (Section 4.2). Terms such as $(\log L_X)^2$ have no physical motivation, invite overfitting, and violate the assumption of common functional form across groups required by the F -statistic derivation.

4.3.5 Design matrix

The design matrix has shape $(n - 1, 6)$, the first observation being unavailable as a response because it has no lag:

$$X_{lag} = \begin{bmatrix} 1 & \log L_X(2) & \Gamma(2) & \log T_{in}(2) & HR(2) & \log F_R(1) \\ 1 & \log L_X(3) & \Gamma(3) & \log T_{in}(3) & HR(3) & \log F_R(2) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \log L_X(n) & \Gamma(n) & \log T_{in}(n) & HR(n) & \log F_R(n-1) \end{bmatrix} \quad (4.7)$$

$$Y = [\log F_R(2), \log F_R(3), \dots, \log F_R(n)]^\top \quad (4.8)$$

with the OLS solution given by the normal equation

$$\hat{w} = (X_{lag}^\top X_{lag})^{-1} X_{lag}^\top Y, \quad \hat{w} = [w_0, w_1, w_2, w_3, w_4, \phi]^\top \quad (4.9)$$

$$\hat{\varepsilon} = Y - X_{lag} \hat{w}, \quad RSS_{global} = \hat{\varepsilon}^\top \hat{\varepsilon} \quad (4.10)$$

RSS_{global} measures how well a **single** causal model fits the entire series and serves as the baseline against which all partitions are compared.

4.4 Gate 1: the lagged Chow test

4.4.1 Hypothesis and intuition

$$H_0 : w_A = w_B \quad (4.11)$$

Rejection indicates that the causal mechanism differs between the two groups. The intuition is as follows: if the hard and soft states obey different causal laws, a model fitted to hard-state observations alone attains a substantially lower residual sum of squares than a pooled model forced to describe both with one coefficient vector. That forced pooling generates excess error, and the test measures whether the excess is significant.

Note that w includes the intercept w_0 ; this has a direct consequence for the design of the validation data, addressed in Section 5.1.

4.4.2 Construction

For a candidate split, let A denote observations before and B after. Separate fits give

$$\hat{w}_A = (X_A^\top X_A)^{-1} X_A^\top Y_A, \quad RSS_A = \|Y_A - X_A \hat{w}_A\|^2 \quad (4.12)$$

$$\hat{w}_B = (X_B^\top X_B)^{-1} X_B^\top Y_B, \quad RSS_B = \|Y_B - X_B \hat{w}_B\|^2 \quad (4.13)$$

The cost of pooling is $RSS_{global} - (RSS_A + RSS_B)$, which is near zero when both groups share one causal law. Normalising by the within-group variance $\hat{\sigma}^2 = (RSS_A + RSS_B)/(n - 2q)$ yields the lagged Chow statistic:

$$F_{lag} = \frac{[RSS_{global} - (RSS_A + RSS_B)]/q}{(RSS_A + RSS_B)/(n - 2q)} \quad (4.14)$$

The numerator is normalised by q because splitting one model into two gains exactly q additional parameters; the statistic is therefore *cost of pooling per additional parameter*. The denominator degrees of freedom $n - 2q$ are those remaining after fitting two models of q parameters each.

4.4.3 Degrees of freedom

The value $q = 6$ follows directly from the structural model of Section 4.3 and is not a tuning parameter: the intercept w_0 (which absorbs $\xi_M \log M_{BH}$), the four spectral-state coefficients w_1 – w_4 , and the autoregressive coefficient ϕ .

4.4.4 Distribution

Under H_0 and Gaussian i.i.d. residuals, the numerator and denominator are independent chi-square variates,

$$[RSS_{global} - (RSS_A + RSS_B)]/\sigma^2 \sim \chi^2(q), \quad (RSS_A + RSS_B)/\sigma^2 \sim \chi^2(n - 2q) \quad (4.15)$$

and the ratio of two independent chi-square variables each divided by its degrees of freedom is F -distributed by definition:

$$F_{lag} \sim F(q, n - 2q) \quad \text{under } H_0 \quad (4.16)$$

$$p = P(F(q, n - 2q) > F_{obs}) = 1 - \text{CDF}_F(F_{obs}, q, n - 2q) \quad (4.17)$$

H_0 is rejected at $\alpha = 0.05$. The Gaussian assumption entering here is load-bearing and constrains the admissible causal discovery methods downstream; this is developed in Section 4.6.

4.5 Locating the boundary and Gate 2

4.5.1 Binary segmentation

The test of Section 4.4 requires a split point to be supplied. Since the boundary location is unknown, the statistic is evaluated at every admissible index and maximised:

$$\tau^* = \arg \max_{\tau} F_{lag}(A(\tau), B(\tau)) \quad (4.18)$$

where $A(\tau) = \{2, \dots, \tau\}$ and $B(\tau) = \{\tau + 1, \dots, n\}$. The maximising index is the location at which the data most strongly prefers two models to one.

Exhaustive search over multi-way partitions is intractable, as the number of ways to partition n observations into K regimes grows exponentially in K . Binary segmentation is a greedy divide-and-conquer approximation: the single best split is located, and the procedure then recurses independently on each child segment. The approximation is not guaranteed globally optimal, a limitation discussed in Section B.1.

A minimum segment size of $2q = 12$ is enforced by two stacking constraints: a segment with fewer than q observations does not permit inversion of $X^\top X$, and the F -test denominator degrees of freedom $n - 2q$ approach zero as $n \rightarrow 2q$, rendering the statistic unstable. This is a hard floor rather than a recommendation; segments near the floor yield noisy coefficient estimates and correspondingly unreliable Stage 2 graphs.

Note on splitting variable. Splitting is performed on the time index only. Feature-threshold splitting of the form (j^*, τ^*) is not employed, as the autoregressive term of (4.5) presupposes temporal contiguity within each segment.

4.5.2 Derivation of the BIC criterion

A significant F -statistic establishes that a detectable difference exists, but binary segmentation is greedy and will, given sufficient noise, always locate some apparently favourable split. Without a stopping rule the recursion terminates only when each observation forms its own segment. A complexity criterion is therefore required, and is derived here from first principles to make explicit how it specialises to the present model.

Step A — the Gaussian likelihood.

For $Y = Xw + \varepsilon$ with $\varepsilon_t \sim N(0, \sigma^2)$ i.i.d.,

$$\log P(Y | w, \sigma^2, M) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{RSS}{2\sigma^2} \quad (4.19)$$

Step B — substitution of the MLE.

With $\hat{\sigma}^2 = RSS/n$,

$$\log P(Y | \hat{w}, \hat{\sigma}^2, M) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log\left(\frac{RSS}{n}\right) - \frac{n}{2} \quad (4.20)$$

The terms $-\frac{n}{2} \log(2\pi)$ and $-\frac{n}{2}$ do not depend on the model and cancel in any comparison, leaving

$$\log P \approx -\frac{n}{2} \log\left(\frac{RSS}{n}\right) \quad (4.21)$$

Step C — the criterion. Schwarz's criterion, obtained from the Laplace approximation to the marginal likelihood, is $BIC = -2 \log P(Y | \hat{\theta}, M) + k \log(n)$ for k free parameters. Substituting (4.19),

$$BIC = n \log\left(\frac{RSS}{n}\right) + k \log(n) \quad (4.22)$$

Step D — the one-model case. For a single global model over n observations with $q = 6$ parameters,

$$BIC_1 = n \log\left(\frac{RSS_{global}}{n}\right) + q \log(n) \quad (4.23)$$

Step E — the two-model case. For disjoint segments the combined log-likelihood is the sum of the two independent log-likelihoods, each with its own MLE $\hat{\sigma}_A^2 = RSS_A/n_A$ and $\hat{\sigma}_B^2 = RSS_B/n_B$. With $2q$ total free parameters,

$$BIC_2 = n_A \log\left(\frac{RSS_A}{n_A}\right) + n_B \log\left(\frac{RSS_B}{n_B}\right) + 2q \log(n) \quad (4.24)$$

Step F — the choice of $\log(n)$. The penalty in (4.22) uses the *total* sample size n rather than the segment sizes. This follows the Bai–Perron convention for structural break models and is necessary rather than cosmetic. Were the penalty written $q \log(n_A) + q \log(n_B)$, it would shrink for small segments, making small

splits progressively cheaper and systematically encouraging over-partitioning — the precise failure the criterion exists to prevent. Using total n in both (4.21) and (4.22) places the two scores on a common scale so that the comparison isolates the fit–complexity trade-off.

Step G — decision rule. A split is accepted if and only if $BIC_2 < BIC_1$. Subtracting (4.21) from (4.22), the net penalty increase from splitting is

$$2q \log(n) - q \log(n) = q \log(n) \quad (4.25)$$

which is the exact price, in units of $-2 \log$ -likelihood, of admitting one additional regime.

4.5.3 Why BIC rather than AIC

AIC imposes a penalty of $2k$, constant in sample size; BIC imposes $k \log(n)$, which grows with the data. Table 4.3 gives the comparison at $k = q = 6$.

Table 4.3: Complexity penalty comparison

n	AIC penalty	BIC penalty	Ratio
800	12	40.1	$3.3\times$
5,000	12	51.1	$4.3\times$
10,000	12	55.3	$4.6\times$

For regime discovery the error costs are asymmetric: a false split places observations in the wrong group and corrupts the downstream causal graph for *both* resulting regimes, whereas a missed split yields a conservative, pooled description. BIC’s heavier and sample-size-growing penalty is therefore the appropriate choice, and the advantage widens with the data volumes characteristic of high-cadence monitoring.

4.5.4 The two-gate design

The two criteria answer different questions and neither is redundant. The Chow test asks whether an observed difference is statistically distinguishable from finite-sample variation; BIC asks whether the resulting model complexity is justified by the improvement in fit. A candidate split may satisfy the first and fail the second. Section 5.3 reports that precisely this occurs at both terminal nodes of the validation run, providing direct empirical evidence that the second gate is load-bearing.

4.6 Identifiability

For a linear structural equation model with Gaussian noise, edge direction within a *chain* ($X \rightarrow Y \rightarrow Z$) or *fork* ($X \leftarrow Y \rightarrow Z$) is not identifiable from observational data: for any true $X \rightarrow Y$ there exists a reversed model with rescaled coefficient and noise variance inducing an identical joint distribution. No algorithm can recover the direction, as the information is absent from the data.

A *collider* — several mutually independent parents converging on a common effect — is the exception. Collider orientation is recoverable from conditional-independence

structure alone, irrespective of noise distribution; this is the v-structure argument on which constraint-based discovery rests generally.

The model of Section 4.3 is a collider. $\log L_X$, Γ , $\log T_{in}$, and HR are specified as parents of $\log F_R$ and not of one another. Direction is therefore identifiable here for structural reasons rather than incidentally.

This has a consequence for possible extensions that is worth stating explicitly. Suppose the model were enriched with causal chains among the spectral variables. Those edges would be chains and forks, and recovering their orientation would require non-Gaussian innovations together with a LiNGAM-family estimator. However, the F -distribution result (4.14) requires Gaussian residuals for the numerator and denominator to be independent chi-square variates, and the BIC derivation (4.17)–(4.20) begins from the Gaussian log-likelihood. Adopting non-Gaussian noise to secure identifiability in Stage 2 would therefore weaken the distributional derivations underpinning both gates of Stage 1. The collider specification avoids this tension: Gaussian noise is retained, (4.14) and (4.20) hold exactly as derived, and Stage 2 orientations remain identifiable. The restriction to a collider structure is thus a defended design decision rather than a simplifying convenience.

4.7 Stage 2: regime-specific causal graphs

Each leaf is a temporally contiguous segment treated as a stable causal regime, and DYNOTEARS is applied independently within each. Two matrices are estimated simultaneously: W , in which $W_{ij} \neq 0$ indicates that variable j affects variable i within the same time step, and A , in which $A_{ij} \neq 0$ indicates that variable j at $t - 1$ affects variable i at t . Acyclicity of the contemporaneous graph is enforced by the smooth algebraic constraint $\text{tr}(e^{W \circ W}) - d = 0$, replacing combinatorial search with gradient-based optimisation.

DYNOTEARS is preferred to the PC algorithm on two grounds. The first, noted in Section 2.1.6, is that PC returns a skeleton and partial orientation without edge weights and requires a number of conditional independence tests that grows sharply once lagged variables are admitted. The second is decisive in practice: the Fisher-Z test assumes independent samples, an assumption violated outright by consecutive observations of an autoregressive process with $\phi \approx 0.9$. Applied to such data the test's null distribution is misspecified, and in preliminary experiments this produced orientations into variables that have no parents by construction.

Two implementation requirements are recorded. First, variables must be standardised within each leaf before fitting, as the DYNOTEARS objective and its sparsity threshold are scale-sensitive; without standardisation, edges belonging to lower-variance variables are systematically underweighted and may be eliminated by the L_1 penalty. Second, the terminality of F_R must be encoded as explicit per-edge exclusions forbidding F_R at any lag from being a parent of L_X , Γ , T_{in} , or HR . A blanket node-level prohibition also suppresses the edge $F_R(t - 1) \rightarrow F_R(t)$, which the model of (4.5) explicitly requires.

4.8 Stage 3: causal effect estimation and validation

Stages 1 and 2 yield a graph; Stage 3 quantifies the effects it encodes and assesses how far those quantities should be trusted.

For each discovered parent of F_R within each regime, the causal effect is estimated by backdoor adjustment, conditioning on the remaining parents. Unpenalised regression is used here, for the reason given in Section 4.3.4: with the structure already determined, regularisation contributes only bias. The lag term $\log F_R(t-1)$ must be included in every adjustment set, since the predictors are autocorrelated with their own past and $F_R(t-1)$ lies downstream of that past; omitting it allows a predictor to absorb part of the lag term's effect.

Three refutation procedures are applied. A **random common cause** test injects an irrelevant synthetic covariate, under which a valid estimate should be unchanged. A **placebo treatment** test replaces the treatment with a permuted version, under which the estimated effect should collapse to zero; this is the diagnostic that would expose a procedure generating effects from noise. A **data subset** test re-estimates on an 80

Sensitivity analysis is then applied in the partial- R^2 parameterisation of Cinelli and Hazlett, which reports a *robustness value*: the minimum proportion of residual variance in both treatment and outcome that an unobserved confounder would need to explain in order to reduce the estimate to zero. Unobserved confounding cannot be excluded from observational data, and Section 1.6 identifies it as a principal challenge for this application; the robustness value converts that acknowledged limitation into a stated numerical threshold against which domain knowledge can be brought to bear.

Finally, **graph falsification** tests whether the conditional-independence implications of the entire recovered DAG are consistent with the data, by comparing its violation count against randomly permuted graphs. Where the preceding tests validate individual effects conditional on the graph, this interrogates the graph itself.

4.9 Parameter summary

The complete parameter specification is given in Table 4.4.

Table 4.4: Complete parameter specification

Parameter	Value	Determined by
Variables log-transformed	F_R, L_X, T_{in}	power-law scaling (Section 4.2)
Variables untransformed	Γ, HR	dimensionless, bounded range (Section 4.2)
q (parameters per model)	6	structural model: w_0, w_1-w_4, ϕ (Section 4.4.3)
Chow numerator d.f.	$q = 6$	parameters gained by splitting (Section 4.4.2)
Chow denominator d.f.	$n - 2q$	residual d.f. after two fits (Section 4.4.2)
α	0.05	conventional; asymmetric error cost handled by Gate 2
BIC penalty, one model	$q \log(n)$	Schwarz criterion (Section 4.5.2)
BIC penalty, two models	$2q \log(n)$	Bai–Perron convention, total n (Section 4.5.2)
Net cost of a split	$q \log(n)$	Eq. (4.23)
Lag order	1	minimum sufficient; higher orders raise q (Section 4.3.3)
Minimum segment size	$2q = 12$	invertibility and F -test stability (Section 4.5.1)
Splitting variable	time index	required by AR term (Section 4.5.1)

Chapter 5

Experimental Validation

5.1 Experimental design

5.1.1 Rationale for synthetic validation

Real observational data provides no ground truth for either regime boundaries or causal structure, and consequently permits no direct measurement of recovery accuracy. Synthetic data generated from a structural causal model with known graphs and known coefficients permits exact evaluation against every output of the framework: regime boundaries against true labels via the Adjusted Rand Index, graph structure against the generating DAG, effect magnitudes against the generating coefficients, and temporal persistence against the true ϕ .

5.1.2 Generating process

Two regimes of 5,000 observations each are generated in temporal order and concatenated without shuffling. The four spectral-state variables follow independent mean-reverting first-order autoregressive processes,

$$X(t) = \mu_X + \varphi(X(t-1) - \mu_X) + \epsilon_X(t), \quad \varphi = 0.90 \quad (5.1)$$

with $\epsilon_X(t) \sim N(0, 0.10^2)$ identically for all four. The mean-reverting form is adopted in place of $X(t) = \varphi X(t-1) + \epsilon(t)$ because the latter has stationary mean zero for $\varphi < 1$, so that physically motivated initial values decay away; (5.1) initialised at $X(0) = \mu_X$ is stationary about μ_X from the first step, preserving physical scale and eliminating any requirement for a burn-in period.

The target variable is generated as

$$\log F_R(t) = \mu_F + \phi(\log F_R(t-1) - \mu_F) + \sum_v \beta_v^{(z)}(X_v(t) - \mu_v) + \epsilon_F(t) \quad (5.2)$$

with $\phi = 0.50$. The predictors enter in centred form and F_R is itself mean-reverting; this construction guarantees $E[\log F_R] = \mu_F$ **for any coefficient vector**, so that the two regimes share identical means in every variable.

5.1.3 Why equal means are necessary for a valid test

This design choice requires explicit justification, as it is what makes the validation meaningful. The Chow null hypothesis (4.10) is $H_0 : w_A = w_B$, and w includes the intercept w_0 . A shift in the mean level of the variables between regimes is absorbed by w_0 , and would therefore produce a Chow rejection *through the intercept* rather than through the slope coefficients that encode the causal mechanism. Under such a design the framework would appear to succeed for a reason substantially weaker than the claim being tested, and — more seriously — any structure-blind clustering method would also succeed, rendering the baseline comparison uninformative. With means equalised, the sole difference between the regimes is which variable carries a non-zero coefficient into $\log F_R$. A rejection can then arise only from genuine mechanism change. Table 5.1 confirms the construction.

Table 5.1: Distributional summary of the generated data

Variable	Hard mean	Soft mean	Hard s.d.	Soft s.d.
$\log L_X$	37.4931	37.4756	0.2332	0.2290
Γ	1.6124	1.5909	0.2347	0.2174
$\log T_{in}$	0.2934	0.3289	0.2188	0.2174
HR	0.7669	0.7320	0.2377	0.2297
$\log F_R$	27.9747	27.9595	0.3803	0.2289

Means agree to within 0.04 in every variable. A residual difference in the variance of $\log F_R$ persists (0.380 against 0.229) and is not removable by construction: it follows necessarily from the larger coefficient magnitudes of the hard state acting on equal-variance inputs. This residual is addressed directly in the baseline comparison of Section 5.3.

5.1.4 Ground-truth structures

Table 5.2: Generating coefficients

Coefficient	Hard state (G_1)	Soft state (G_2)
β_{L_X}	+0.70	+0.18
β_{Γ}	−0.30	+0.20
$\beta_{T_{in}}$	0.00 (<i>isolated</i>)	−0.40
β_{HR}	−0.40	0.00 (<i>isolated</i>)
ϕ	+0.50	+0.50

The hypothesised physical interpretation is that the hard state is jet-dominated, with radio emission driven by X-ray luminosity, photon index, and hardness ratio while the inner disk temperature remains causally isolated from the jet mechanism; and that the soft state is disk-dominated, with the hardness ratio isolated in its place.

The generating coefficients are listed in Table 5.2.

The corresponding ground-truth causal structures are shown in Figure 5.1, and the generated series and its true and discovered boundaries are shown in Figure 5.2.

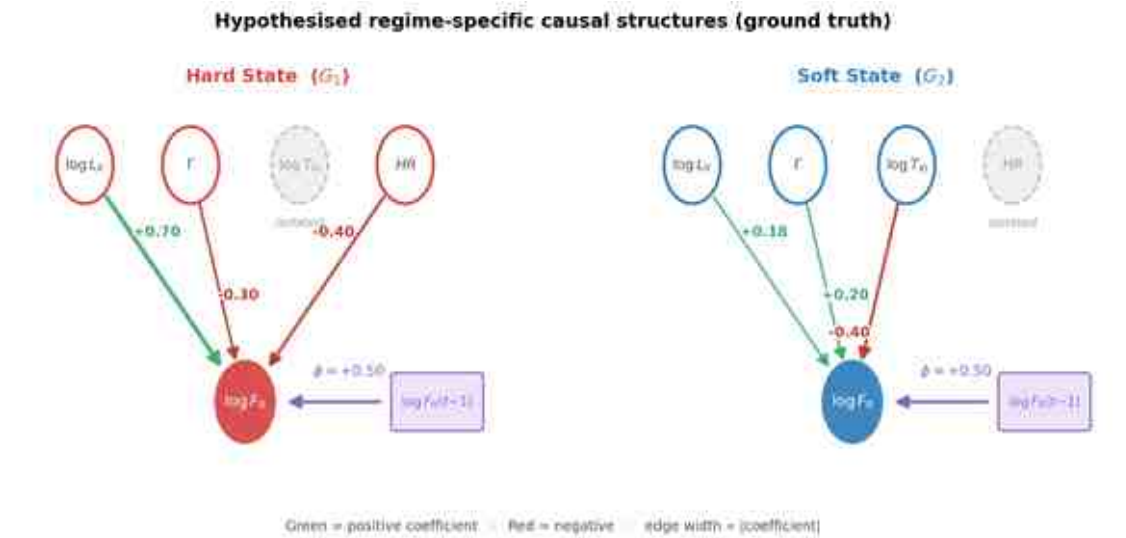


Figure 5.1: Hypothesised regime-specific causal structures. Edge width is proportional to coefficient magnitude; the causally isolated variable in each regime is shown dashed.

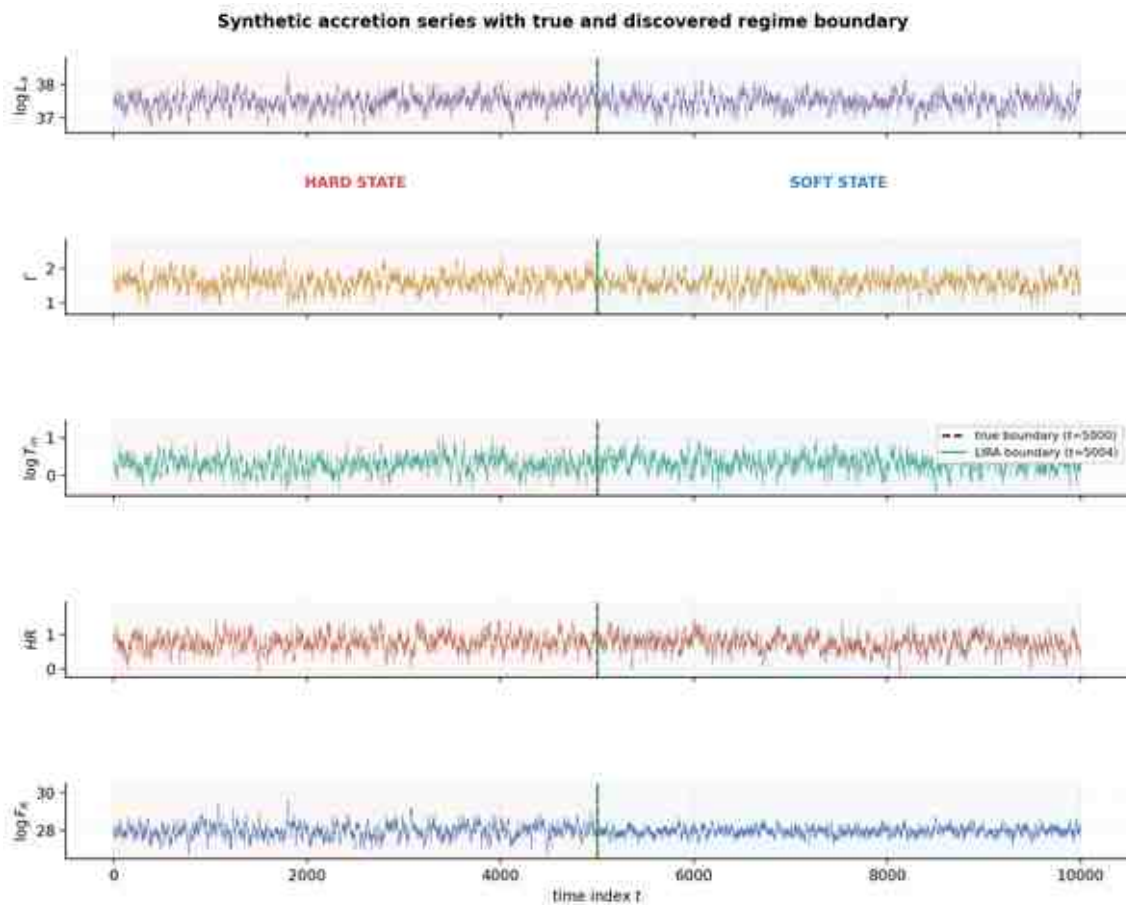


Figure 5.2: The generated series with the true boundary (dashed) and the boundary discovered by LIRA (solid). No level shift is visible at the boundary in any variable, by construction.

5.2 Validation of the autoregressive extension

Section 4.3.3 argues that the lag term is required for the Chow F derivation to hold. This is verified directly. Fitting the static specification (4.4) within the first regime yields residuals with first-order autocorrelation 0.507 and a Ljung–Box statistic at 10 lags of 1713.4 ($p < 10^{-16}$), decisively rejecting independence. Fitting the autoregressive specification (4.5) to the same data reduces the first-order residual autocorrelation to -0.011 , with a Ljung–Box statistic of 13.2 ($p = 0.211$), consistent with independence at any conventional level.

The residual-independence diagnostics are summarised in Table 5.3.

Table 5.3: Residual independence diagnostics (hard-state segment)

Specification	ACF(1)	Ljung–Box(10)	p	Residuals i.i.d.?
Static, Eq. (4.4)	0.507	1713.4	$< 10^{-16}$	rejected
Autoregressive, Eq. (4.5)	-0.011	13.2	0.211	not rejected

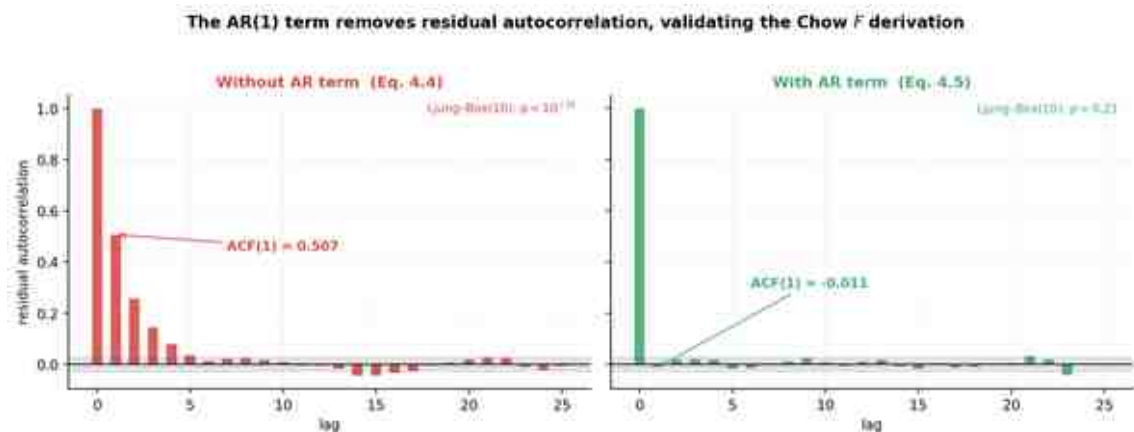


Figure 5.3: Residual autocorrelation functions with and without the AR(1) term. Shaded band gives the 95% interval under independence.

The distributional assumption underlying (4.14) is therefore satisfied in the autoregressive specification and violated in its absence, confirming that the extension is necessary rather than merely beneficial.

Figure 5.3 shows the corresponding residual autocorrelation diagnostics.

5.3 Regime discovery

5.3.1 Boundary localisation

Figure 5.4 shows the lagged Chow statistic evaluated across all admissible split indices at the root node. The statistic rises monotonically toward the true boundary and attains its maximum of $F_{lag} = 963.4$ at $\tau^* = 5004$, against a true boundary at $t = 5000$ — a localisation error of four observations, or 0.04% of the series length. The associated p -value is below the limit of double-precision representation.

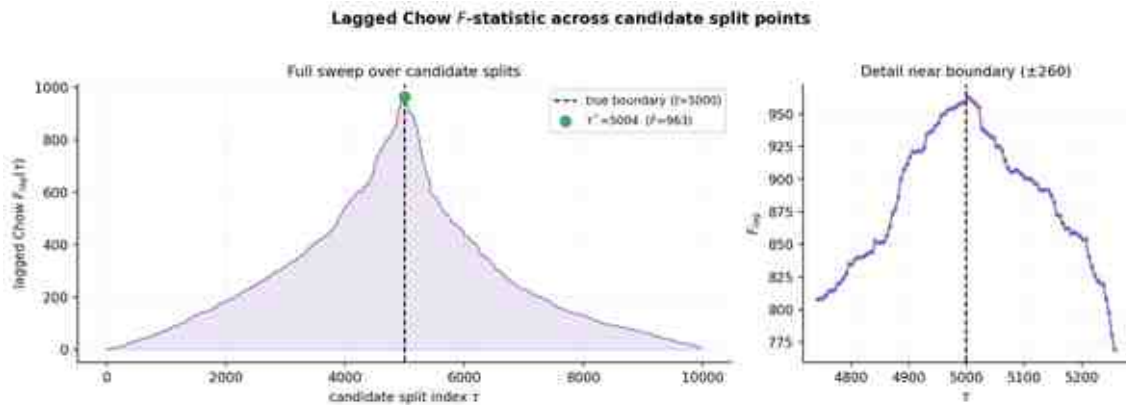


Figure 5.4: Lagged Chow F -statistic across candidate split points, with detail near the boundary.

5.3.2 Operation of the two gates

At the root node both gates are satisfied: $F_{lag} = 963.4$ with $p < 10^{-16}$, and $BIC_1 = -41362$ against $BIC_2 = -45874$, so the split reduces BIC by 4512 and is accepted. Recursion then proceeds into each child.

At both children the behaviour is instructive and directly supports the two-gate design of Section 4.5.4. The Chow test at each child returns a *significant* result — $F_{lag} = 3.15$, $p = 0.004$ on the left and $F_{lag} = 3.27$, $p = 0.003$ on the right — and a single-gate procedure would therefore have partitioned each true regime further, producing spurious boundaries. The BIC gate rejects both: $BIC_1 = -23026$ against $BIC_2 = -22994$ on the left, and $BIC_1 = -22851$ against $BIC_2 = -22820$ on the right, in each case the split increasing BIC by approximately 31 and thus failing (4.23). Recursion terminates with $K = 2$.

The gate decisions at the root and child nodes are summarised in Table 5.4.

Table 5.4: Gate evaluation at each recursion node

Node	n	F_{lag}	p	Gate 1	BIC_1	BIC_2	Gate 2	Outcome
root	9,999	963.38	$< 10^{-16}$	pass	-41362	-45874	pass	split at $t = 5004$
left child	5,003	3.15	0.0044	pass	-23026	-22994	fail	leaf
right child	4,995	3.27	0.0033	pass	-22851	-22820	fail	leaf

That the Chow test alone would have over-partitioned at both terminal nodes is direct empirical evidence that the complexity criterion is load-bearing rather than a formality. This is the clearest available demonstration that the two gates answer genuinely different questions.

Figures 5.5 and 5.6 visualise the BIC decisions and the realised recursion tree, respectively.

5.3.3 Comparison against structure-blind baselines

$K = 2$ regimes are recovered with an Adjusted Rand Index of 0.9984, without K being supplied — it emerges from the two-gate recursion of Section 4.5.4. Table 5.5 compares this against k -means clustering, with the true number of regimes given to

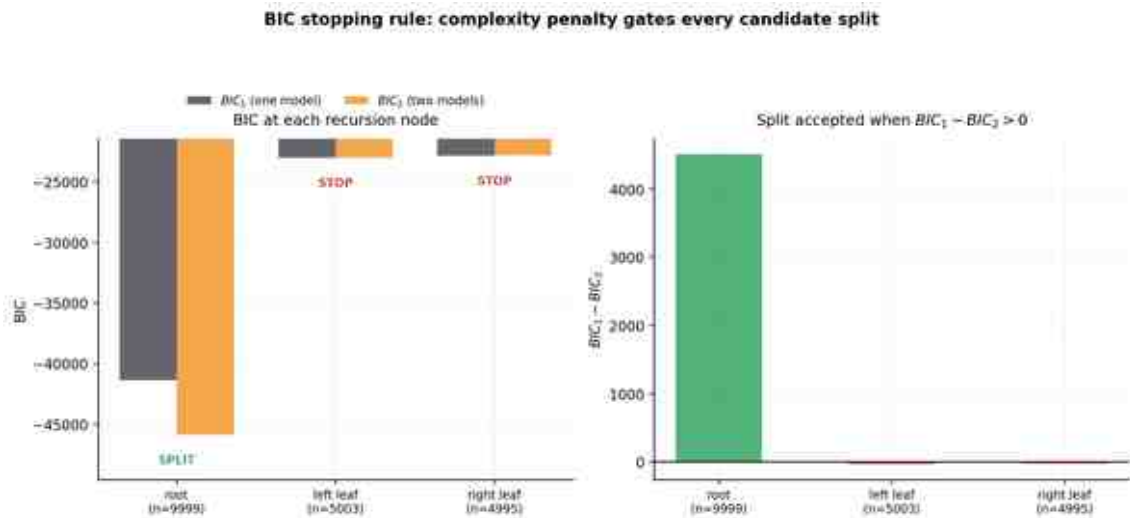


Figure 5.5: BIC evaluation at each recursion node.

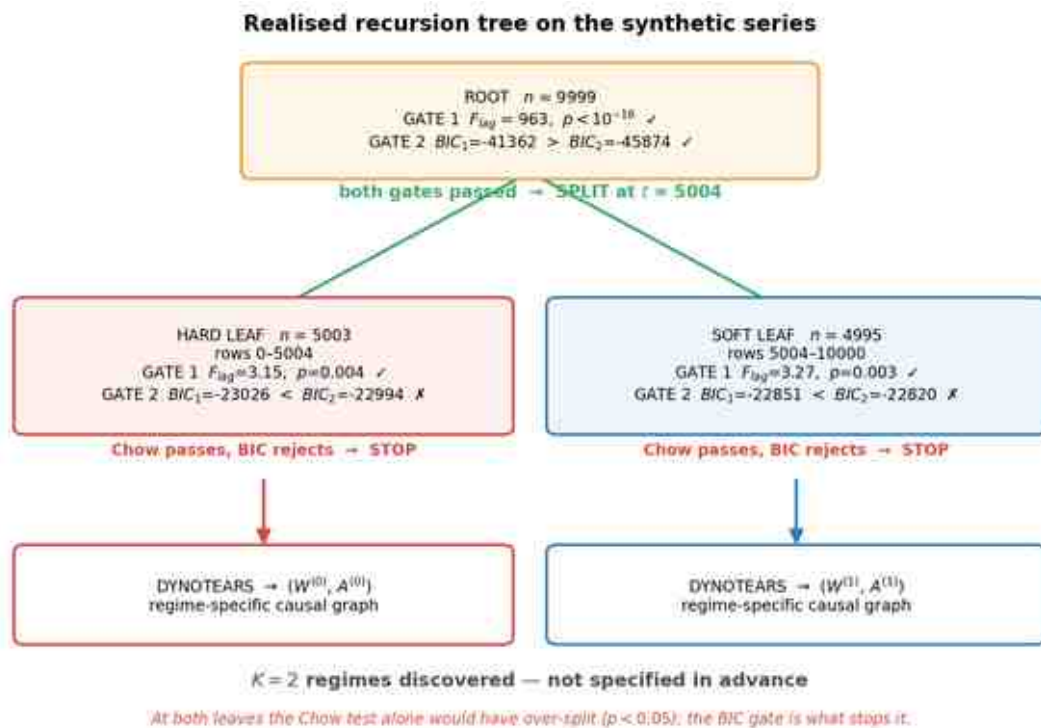


Figure 5.6: Realised recursion tree on the synthetic validation series.

k -means directly (i.e. k -means is run with `n_clusters=2` set explicitly, an advantage LIRA is not given). Even so, feature-space clustering fails to recover the partition in three of the four variants tested.

Table 5.5: Regime recovery accuracy

Method	Basis for grouping	ARI
LIRA (this work)	causal mechanism invariance, K not supplied	0.9984
k -means, raw features ($K = 2$ given)	feature-space proximity	0.0011
k -means, standardised features ($K = 2$ given)	feature-space proximity	0.0009
k -means, rolling means ($K = 2$ given, $w = 200$)	local level	0.0112
k -means, rolling s.d. of $\log F_R$ ($K = 2$ given, $w = 200$)	local dispersion	0.5867

Three of the four baselines fail almost completely, even with the correct number of regimes given directly — consistent with the design intent of Section 5.1.3 that the regimes are not separable by distributional similarity.

The dispersion-based baseline attains $\text{ARI} = 0.587$ and is reported rather than omitted because it quantifies a genuine residual leakage in the experimental design. As noted in Section 5.1.3, the variance of $\log F_R$ necessarily differs between regimes because the hard-state coefficients are larger in magnitude. A baseline constructed specifically to monitor that one statistic recovers part of the partition even so; it nonetheless recovers substantially less than LIRA, and required the correct K to be supplied in advance, a step LIRA does not require.

Figure 5.7 shows the regime-recovery comparison against the structure-blind baselines.

5.4 Structural coefficient recovery

Fitting (4.5) within each discovered leaf recovers the generating coefficients to high accuracy in both regimes, including correct identification of the causally isolated variable in each.

The recovered structural coefficients are reported in Table 5.6.

Table 5.6: Recovered structural coefficients

Coefficient	Hard state			Soft state		
	True	Recovered	Error	True	Recovered	Error
β_{L_X}	+0.70	+0.7172	+0.017	+0.18	+0.1830	+0.003
β_{Γ}	−0.30	−0.2989	+0.001	+0.20	+0.2030	+0.003
$\beta_{T_{in}}$	0.00	+0.0049	+0.005	−0.40	−0.3894	+0.011
β_{HR}	−0.40	−0.4015	−0.002	0.00	−0.0070	−0.007
ϕ	+0.50	+0.4931	−0.007	+0.50	+0.4958	−0.004

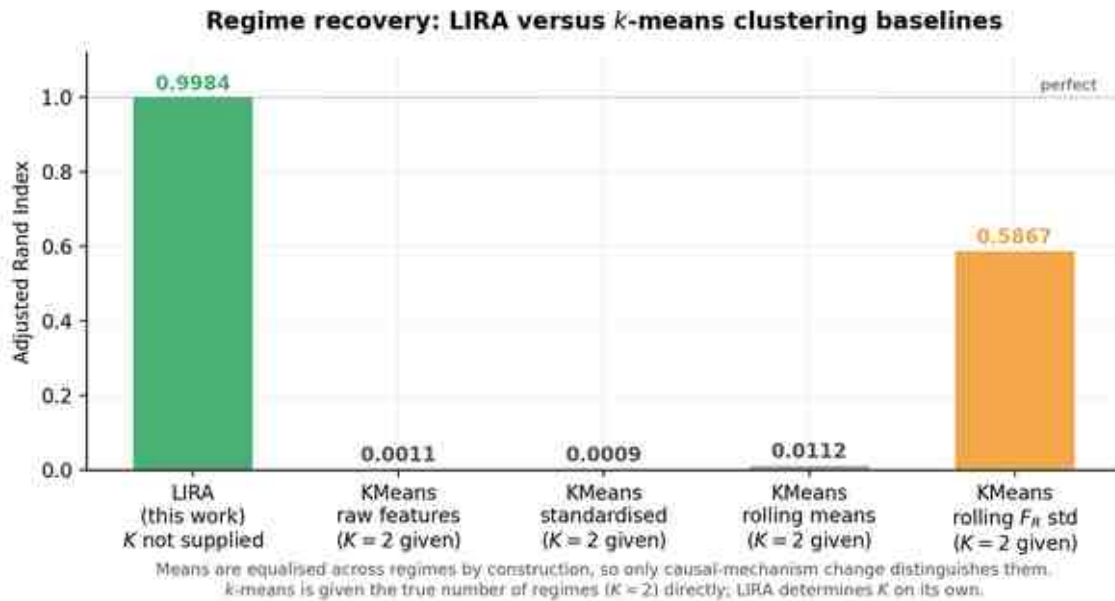


Figure 5.7: Regime recovery: LIRA against structure-blind clustering baselines.

The isolated variable is recovered as $+0.005$ in the hard state and -0.007 in the soft state, in both cases within one hundredth of zero and an order of magnitude smaller than any active coefficient. The autoregressive coefficient is recovered within 0.007 of its true value in both regimes.

Temporal persistence. The recovered ϕ values of 0.493 and 0.496 are statistically indistinguishable, correctly reflecting that the generating process assigns identical persistence to both regimes. The framework therefore does not manufacture a persistence difference where none exists. It should be noted that the abstract’s claim regarding differential temporal persistence across accretion states is a hypothesis about *real* systems; the present synthetic design does not encode such a difference, and testing it requires either a generator in which ϕ varies by regime or application to observational data.

5.4.1 The pooled-model baseline

To assess whether segmentation is necessary at all, (4.5) is fitted once to the unsegmented series. The result is a structure corresponding to neither regime. The pooled and regime-specific estimates are compared in Table 5.7.

Table 5.7: Pooled model versus regime-specific models

Coefficient	True (Hard)	True (Soft)	Pooled fit
β_{L_X}	+0.70	+0.18	+0.2504
β_{Γ}	−0.30	+0.20	−0.0285
$\beta_{T_{in}}$	0.00	−0.40	−0.1167
β_{HR}	−0.40	0.00	−0.1170
ϕ	+0.50	+0.50	+0.7650

Every coefficient is distorted. The coefficient on Γ , which is strongly negative in

one regime and positive in the other, is averaged almost to zero (-0.029), implying no relationship where two strong and opposite relationships exist. Both isolated variables acquire spurious non-zero coefficients. The autoregressive coefficient is inflated by more than 50%, from a true 0.50 to 0.765, as the pooled model attributes unexplained between-regime structure to temporal persistence.

This quantifies the claim made in Section 1.2 that fitting a single pooled model across physically distinct accretion states produces biased estimates and obscures the true coupling laws. It is the most direct available justification for the framework. The recovered and pooled coefficients are compared visually in Figure 5.8.

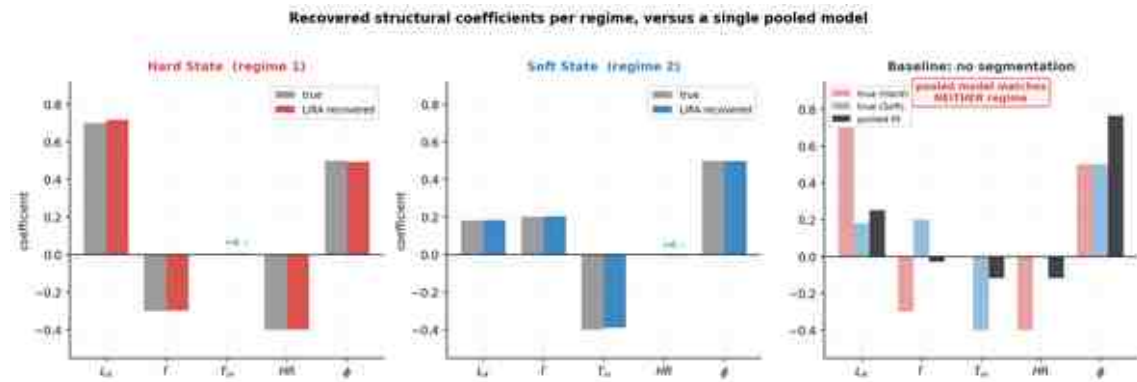


Figure 5.8: Recovered coefficients per regime, and the pooled model that matches neither.

5.5 Regime-specific causal graph recovery

DYNOTEARS applied independently within each discovered leaf recovers the correct graph structure in both regimes.

The contemporaneous and lagged edge estimates are reported in Tables 5.8 and 5.9, respectively.

Table 5.8: Contemporaneous edge recovery ($\hat{W}$)

Edge	Hard: true	Hard: $\hat{W}$	Soft: true	Soft: $\hat{W}$
$L_X \rightarrow F_R$	present (+)	+0.315 yes	present (+)	+0.079 yes
$\Gamma \rightarrow F_R$	present (−)	−0.100 yes	present (+)	+0.090 yes
$T_{in} \rightarrow F_R$	absent	absent yes	present (−)	−0.212 yes
$HR \rightarrow F_R$	present (−)	−0.157 yes	absent	absent yes
any edge into L_X	absent	absent yes	absent	absent yes

Every edge sign is correct, the isolated variable is correctly excluded in each regime, and the root variable L_X acquires no parent other than its own lag — the clearest single diagnostic, since L_X is the one variable with no causal parents by construction. Magnitudes are systematically attenuated relative to the generating coefficients, as expected from L_1 soft-thresholding, which shrinks each estimate by approximately λ ; the structural conclusions are unaffected, and unbiased magnitudes are obtained separately in Section 5.6. Figure 5.9 compares the recovered adjacency matrices with the generating contemporaneous and lagged structures.

Table 5.9: Lagged edge recovery ($\hat{A}$, self-lags)

Self-lag	True	Hard: $\hat{A}$	Soft: $\hat{A}$
L_X	0.90	0.851	0.851
Γ	0.90	0.854	0.839
T_{in}	0.90	0.843	0.838
HR	0.90	0.854	0.847
F_R	0.50	0.527	0.495

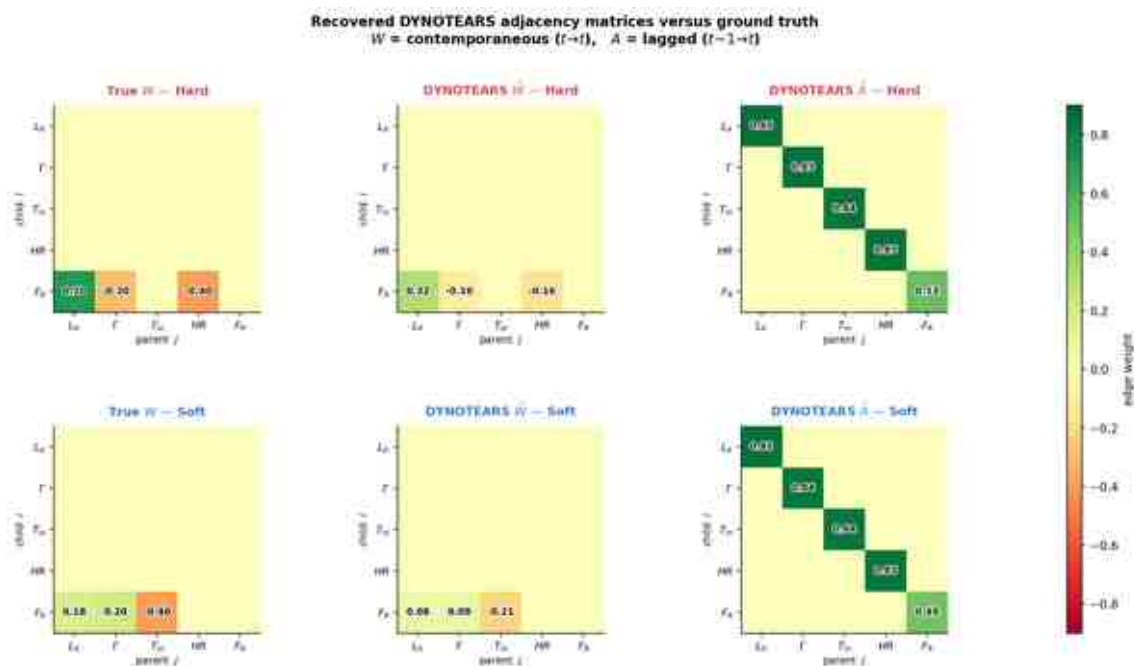


Figure 5.9: Recovered adjacency matrices against ground truth, for both the contemporaneous and lagged components.

5.6 Causal effect estimation and validation

5.6.1 Effect estimates

Estimating each effect by backdoor adjustment with the remaining parents in the adjustment set recovers the generating coefficients substantially more accurately than the penalised Stage 2 weights, as anticipated in Section 4.8. The estimated causal effects are listed in Table 5.10.

Table 5.10: Estimated causal effects, refutation outcomes, and robustness

Regime	Treatment	Estimate	True	Random c.c.	Placebo	Subset	Robustness
Hard	L_X	+0.7111	+0.70	+0.7110	+0.0016	+0.7116	+0.680
Hard	Γ	-0.2964	-0.30	-0.2964	-0.0003	-0.2963	+0.473
Hard	HR	-0.3981	-0.40	-0.3981	+0.0011	-0.3985	+0.556
Hard	$F_R(t-1)$	+0.4983	+0.50	+0.4983	-0.0006	+0.4979	+0.734
Soft	L_X	+0.1799	+0.18	+0.1800	-0.0006	+0.1800	+0.325
Soft	Γ	+0.1994	+0.20	+0.1993	-0.0005	+0.1999	+0.338
Soft	T_{in}	-0.3833	-0.40	-0.3833	-0.0001	-0.3829	+0.515
Soft	$F_R(t-1)$	+0.5062	+0.50	+0.5063	-0.0007	+0.5059	+0.728

Every estimate lies within 0.017 of its generating value. All three refutation tests behave as required in every case: the random common cause and data subset procedures leave estimates unchanged to three or four decimal places, and the placebo treatment collapses every effect to within 0.002 of zero.

The effect estimates and refutation outcomes are visualised in Figure 5.10.

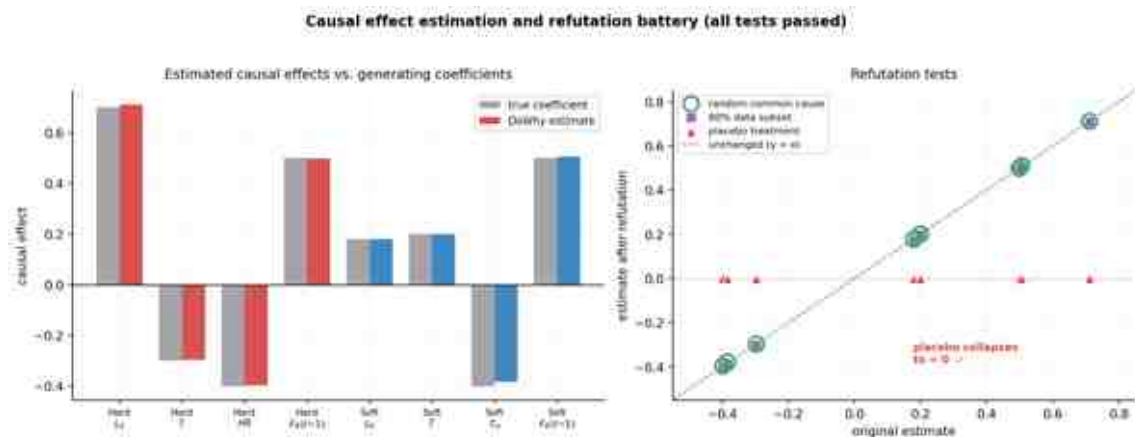


Figure 5.10: Causal effect estimation and refutation battery; all reported tests pass.

5.6.2 Sensitivity to unobserved confounding

The robustness value from the Cinelli–Hazlett sensitivity analysis quantifies the minimum residual variance an unobserved confounder would need to explain in both the treatment and outcome to reduce an estimated effect to zero. Figure 5.11 shows

that robustness increases with the magnitude of the estimated effect, with a Pearson correlation of $r = 0.888$. The weaker couplings in the soft state therefore require less hidden confounding to be explained away and warrant correspondingly more careful causal interpretation than the stronger hard-state effects.

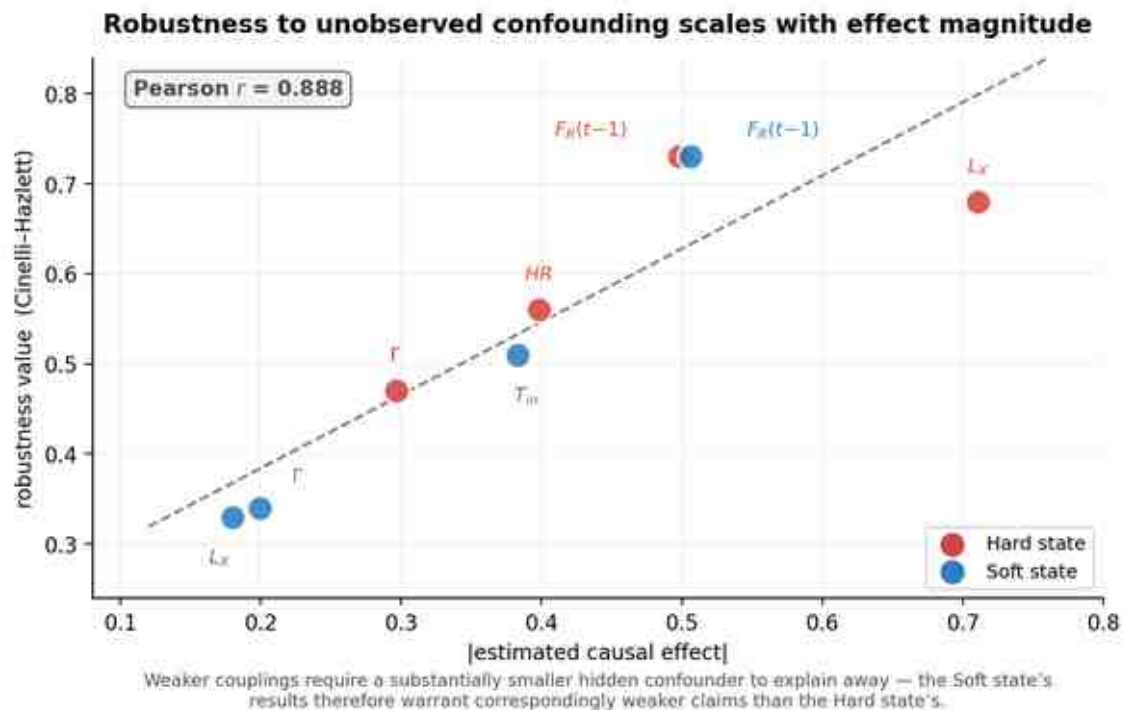


Figure 5.11: Robustness to unobserved confounding as a function of estimated causal-effect magnitude.

5.6.3 Graph falsification

Falsification testing compares the conditional-independence implications of each recovered DAG against randomly permuted graphs. At 20 permutations, both regimes return an informative verdict and neither DAG is rejected at $\alpha = 0.05$: one of 20 permutations falls within the Markov equivalence class in each case, with the hard-state graph violating 15 of 22 local Markov conditions and the soft-state graph 10 of 22, each outperforming 95% of permuted alternatives. The falsification results are summarised in Table 5.11.

Table 5.11: Graph falsification summary

Regime	LMC violations	Better than	Permutations in MEC	p	Verdict
Hard	15 / 22	95.0%	1 / 20	0.05	not rejected
Soft	10 / 22	95.0%	1 / 20	0.05	not rejected

Limitation. With 20 permutations the p -value is quantised to multiples of 0.05 and can therefore only report values at the decision threshold. The verdict is consequently reported as indicative rather than conclusive, and re-running at 100 or more permutations is identified as required before publication.

5.7 Summary of validation outcomes

Table 5.12 consolidates the main validation outcomes.

Table 5.12: Summary of experimental findings

Claim	Evidence	Outcome
AR(1) term restores residual independence	Ljung–Box p : $< 10^{-16} \rightarrow 0.211$	confirmed (Section 5.2)
Boundary is localised accurately	$\tau^* = 5004$ vs. true 5000	confirmed (Section 5.3.1)
K is determined automatically	$K = 2$ recovered, not specified	confirmed (Section 5.3.2)
Both gates are necessary	Chow passes at both leaves; BIC rejects	confirmed (Section 5.3.2)
Regimes are not separable by feature distribution	k -means (given $K = 2$) ARI ≤ 0.012 for three of four variants versus LIRA 0.998	confirmed (Section 5.3.3)
Coefficients are recovered accurately	Maximum error 0.017 across both regimes	confirmed (Section 5.4)
Pooling produces a spurious structure	β_{Γ} averaged to -0.029 ; ϕ inflated to 0.765	confirmed (Section 5.4.1)
Regime-specific graphs are recovered	All signs correct; isolated variable excluded in each	confirmed (Section 5.5)
Effects survive refutation	Placebo < 0.002 ; other estimates unchanged	confirmed (Section 5.6.1)
Robustness scales with effect size	$r = 0.888$	confirmed (Section 5.6.2)
Recovered graphs are not falsified	Neither rejected at $\alpha = 0.05$	indicative only (Section 5.6.3)

Chapter 6

Conclusion

This thesis has introduced LIRA (Latent Invariant Regime Analysis), a self-supervised causal framework for discovering latent causal regimes and their regime-specific causal structures directly from unlabelled black hole accretion time series. Motivated by the power-law physics of the Fundamental Plane of Black Hole Activity, the framework models accretion observables within a log-linear structural causal specification in which black hole mass is absorbed into the intercept and temporal state variation is carried by four spectral-state diagnostics. Regime boundaries are located by greedy recursive partitioning driven by a first-order autoregressive Chow F -test, and are subject to an independent complexity criterion derived from the Gaussian likelihood, which determines the number of regimes automatically. Within each discovered regime, DYNOTEARS recovers contemporaneous and lagged causal structure, and the resulting effects are subjected to refutation testing, sensitivity analysis, and graph falsification.

Validation on synthetic accretion data constructed so that the regimes are distinguishable by causal mechanism alone establishes that the framework recovers the true boundary to within four observations of ten thousand, determines the correct number of regimes without prior specification, and recovers every structural coefficient to within 0.017 of its generating value, including the correct identification of the causally isolated variable in each state. Structure-blind clustering baselines fail almost entirely on the same data, and a pooled model fitted without segmentation recovers a structure corresponding to neither regime – averaging a coefficient that is strongly negative in one state and positive in the other almost to zero, and inflating the persistence parameter by more than half. The two-gate design is shown to be load-bearing rather than redundant: at both terminal nodes the invariance test alone would have partitioned further, and only the complexity criterion prevents it. Three limitations are acknowledged. Empirical validation to date rests on synthetic rather than observational data; the linearity and piecewise-stationarity assumptions approximate what are in reality nonlinear and continuously evolving accretion dynamics; and the identifiability argument advanced in Section 4.6 holds for the collider specification adopted here and would not extend without modification to a model containing causal chains among the spectral variables. The sensitivity analysis reported in Section 5.6.2 addresses the hidden-confounding concern raised at the outset not by excluding it, which observational data cannot do, but by quantifying how strong an unmeasured confounder would need to be – and in doing so establishes that the soft state’s weaker coupling warrants correspondingly weaker causal

claims than the hard state's.

The immediate direction for future work is application to real observational data from the WATCHDOG catalogue and to well-monitored sources such as GRS 1915+105 and Cygnus X-1, where the discovered regimes may be compared against expert spectral-state classifications. Beyond this, extension to nonlinear intra-regime mechanisms, comparison against CASTOR on a common benchmark, and relaxation of the piecewise-constant assumption in the direction of continuously evolving structure represent the natural continuations of this work.

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